NOT MEASUREMENT SENSITIVE

MIL-PRF-87269A
1 October 1995

SUPERSEDING
MIL-D-87269
20 November 1992

PERFORMANCE SPECIFICATION

DATA BASE, REVISABLE - INTERACTIVE ELECTRONIC TECHNICAL MANUALS, FOR THE SUPPORT OF

This performance specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This performance specification prescribes the requirements for an Interactive Electronic Technical Manual Data Base (IETMDB) to be constructed by a weapon-system contractor for the purpose of creating Interactive Electronic Technical Manuals (IETM). The requirements herein cover the specification for the IETMDB and are intended to apply to one or both of two modes as specified in a contract: (1) the interchange format for the data base to be delivered to the Government; or (2) the structure and the naming of the elements of the data base created and maintained by the contractor for purposes of creating IETMs which are in turn delivered to the Government.

1.2 Paragraphs with limited applicability. This specification contains paragraphs and specific requirements which are applicable to all Services. Such paragraphs or requirements are prefixed to indicate the Services to which they pertain: (A) for Army; (N) for Navy; (M) for Marines; and (F) for Air Force.

2. APPLICABLE DOCUMENTS.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Det 2 HQ ESC/AV-2, 4027 Col Glenn Hwy, Suite 300 Dayton, OH 45431-1672; by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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2.1 General. The Documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-PRF-87268	-	Manual, Technical - General Content, Style, Format, and User Requirements for Interactive Electronic Technical Manuals

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-1840	-	Automated Interchange of Technical Information

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATIONS

DEPARTMENT OF DEFENSE

DOD 5200.1-R	-	Information Security Program Regulations
DOD 5220.22-M	-	Industrial Security Manual for Safeguarding Classified Information

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, D.C. 20402)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

ISO 8879	-	Information Processing - Text and Office Systems - Standard Generalized Markup Language (SGML)
ISO 10744	-	Information Technology - Hypermedia/Time-Based Structuring Language (HyTime)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 General requirements. An IETMDB developed in accordance with this specification shall conform to the Content Data Model (CDM) specified herein. The CDM employs a two layered approach to define technical information (TI). The top layer, called the "Generic Layer", shall define the semantic rules for the data characteristics. The generic layer is defined in Appendices A and C of this specification. The bottom layer, called the "Content Specific Layer", shall employ the generic layer when defining elements for weapon system specific TI. Appendices B and D contain a content specific layer model developed for organizational level maintenance. Many content specific layers can be developed in accordance with the generic layer. The CDM generic layer defined in Appendices A and C of this specification are the DoD standard for any IETM technical information, data base procured using this specification. In addition, unless otherwise specified by the procuring activity, the content specific layer Document Type Definition (DTD) defined in Appendices B and D shall also be part of this specification (see 6.2). If Appendices B and D of this specification are not specified by the procuring activity, some other content specific layer DTD shall be specified and approved by the government. The IETMDB can be invoked by a procuring activity in either one of two modes as follows, depending on whether a data base is (1) specified for interchange and delivery to the Government, or (2) being developed and maintained for the subsequent preparation of IETMs, but not actually delivered to the Government.

3.1.1 Data Base interchange requirements. When specified, IETMDBs which are to be delivered to the Government under this specification shall be structured and tagged in accordance with the DTDs and the tag set descriptions included as Appendices A through D of this specification (see 6.2).

3.1.2 Data base structuring and data element naming requirements. Unless otherwise specified, a deliverable instance created under this specification shall be structured in accordance with the hierarchical relationships defined in the CDM DTDs contained in Appendices A and B, and created

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and named using the tag set descriptions contained in Appendices C and D (see 6.2). When a tagged instance is not specified for delivery, the contractor shall maintain the ability to map the internal element names to the specified content specific DTD names.

3.2 Format free technical information. The IETMDB shall consist of an assemblage of data elements, including a listing of the specific attributes possessed by the data elements; and a list of explicit relationships providing logical links among the data elements. The relationships incorporated into the data base, by the IETMDB author, shall provide the basis of the technical structure of the IETMs and other logistic support TI which will be extracted from it. The IETMDB shall not contain format directions in the sense of arrangement of text and graphics on a display screen for presentation to the end user. The IETMDB itself shall require a "format" (data base structure) but this specification does not impose structural requirements on the actual Data Base Management System (DBMS) methodology to be employed (i.e., the data base may be either relational or object oriented). The exterior view of the data base to be used for updating, adding cross references, producing tagged output files, etc. shall conform to requirements of this specification.

3.2.1 Data portability. Formatting requirements shall be eliminated from the IETMDB to reduce the overall magnitude of data base and data interchange standardization effort. This shall also permit the use of a less complex DBMS by the contractor which is, in turn, less expensive and easier to modify. The "format-free" nature of the IETMDB shall provide the Government the capability to:

a. Acquire or access the data in a variety of ways (IETMs, other types of logistics reports, training TI, etc.).

b. Subsequently format and style the data in a variety of ways for electronic display options.

3.2.2 Integration support. IETMDBs shall provide direct, on-line data access to a variety of users and to a number of automated logistic support and management information systems throughout the services. Establishment of standard identifiers, data entity relationships, and multiple path access routes to individual data elements shall be part of the IETMDB design and construction.

3.2.3 Data maintainability. The IETMDB shall be constructed with provisions that allows incorporation of any change to automatically update all aspects of the data base affected by that change. This data-maintainability requirement shall involve the following two kinds of changes to the IETMDB:

a. Additions to, eliminations of, or changes to individual data elements and attributes.

b. Changes to relationships including establishment of new relationships or elimination of old relationships.

3.2.4 Additional content specific DTDs. When specified, additional content specific DTDs shall be used in addition to or instead of the content specific DTD defined in Appendices B and D of this specification (see 6.2). These DTDs shall be incorporated into the overall CDM in accordance with the requirements of 3.2.

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3.3 Generic layer. The generic layer of the CDM is defined in the DTD listed in Appendix A. This DTD provides templates, which shall be used to define content specific elements. The generic layer includes a definition for each template and the attribute lists associated with the template. The DTD provides a definition of three other data types; primitive data elements that shall remain standard across all content specific applications; user interaction elements, called dialogs; and the context filtering elements, which shall be used to provide the most appropriate information to a user. The following paragraphs describe the components of the generic layer:

3.3.1 Templates. Templates shall be used as described in Appendix A to define elements declared in content specific DTDs. The generic layer shall contain five templates: Node, Node Alternatives, Node Sequence, If Node, and Loop Node. Each template shall have two components: (1) a set of semantic rules that govern the template's activities, and (2) a list of attributes.

3.3.1.1 Node template. All elements conforming to the node template shall provide the capability for creating composite structures within the content specific layer. Composite structures shall contain primitives, links, and preconditions. When a composite structure contains other composite structures within its content model, this implies hierarchy. Elements employing the node template shall have a set of required attributes as follows.

3.3.1.1.1 Name. The "name" attribute of the element shall consist of the standard nomenclature for an instance of the element.

3.3.1.1.2 Item-Id. The "Item-Id" attribute shall specify the reference designator(s) and other identifiable designator(s) of the system(s), subassemblies, or part(s) referred to by the element.

3.3.1.1.3 Type. The "type" attribute shall specify the type of information contained in the element.

3.3.1.1.4 CDM. The "cdm" attribute shall identify the type of template being employed by the content specific element.

3.3.1.1.5 Ref. The "ref" attribute shall facilitate the reduction of data redundancy by allowing data elements to be referenced.

3.3.1.2 Node Alternatives (Alts) template. All elements conforming to the node alts template shall contain a list of mutually exclusive nodes, only one of which shall be used at the time of presentation.

3.3.1.3 Node Sequence (Seq) template. All elements shall conform to the node seq template group elements together and provide an order or presentation sequence to the elements. The elements conforming to the Node Seq shall also allow an author to define branching logic within the TI.

3.3.1.4 If node template. Elements conforming to the if node template shall provide a method for conditional branching. These elements shall use the same logic as the IF-THEN-ELSE statement in a programming language. The "IF" part is the expression in the content model. The "THEN" part is the first node seq and is selected when the expression evaluates to true. The "ELSE" part is the second node seq, which is optional in the CDM, and is selected when the expression evaluates to not true.

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3.3.1.5 Loop node template. The loop node template shall provide the equivalent of a loop in a programming language. This element shall provide the capability to create either a "FOR" loop or a "WHILE" loop within the data. The expressions and assertions shall be developed in accordance with this template and provide the testing criteria for the loop. The node sequence shall contain the actual elements to be repeated within the loop.

3.3.2 Relational links. Elements shall have relationships to other elements in the TI, when applicable. These relationships shall be represented through two or more link ends. The link element shall provide the capability to show the relationship between several elements. The contractor shall include the specific cross-references to elements within the IETMDB as well as information sources outside the IETMDB.

3.3.2.1 Links to reduce redundancy. Links shall be used to reduce the number of redundant elements by referencing common elements. The templates defined within the generic layer CDM DTD shall define attributes to reduce redundant elements. These elements shall utilize the Standard Generalized Markup Language (SGML) #CONREF reference capability in accordance with International Organization for Standardization (ISO) 8879. The #CONREF attribute shall contain the unique identifier of an element using a template or a location element.

3.3.2.2 Location elements. Location elements are defined by ISO 10744. Elements shall be referenced by other elements in accordance with ISO 10744.

3.3.2.3 Logistics support and task-analysis link. The contractor shall establish linkages (information-access capabilities) with the IETMDB when external logistics support and task-analysis systems have been developed.

3.3.3 Primitive elements. An IETMDB shall be composed of the primitive elements defined in the generic layer DTD. Content and style for these elements shall conform with the requirements of MIL-PRF-87268.

3.3.3.1 Textual information. Textual information shall consist of alphanumeric (i.e., character) data. When required, textual information shall contain embedded references to some higher level elements, such as those describing parts or consumables.

3.3.3.2 Tables. Tables shall be represented as a series of separate entries, each entry being associated with a specific row and column intersection (cell) of a table. Each entry in the table may be associated with other primitive types of information presentation and attributes. Each entry may refer (through a relationship) to any other template element or primitive element in the IETMDB.

3.3.3.3 Graphics. Graphics (drawings, illustrations) information shall be structured in a hierarchical manner and consist of logically related groups. Graphics shall be composed of a series of illustrations which can be overlaid on each other to build a complete graphic. These graphic "building blocks" are called graphic primitives. Graphic primitives may be combined to produce composite information which can be referenced and selected. Graphics shall be composed of information represented in accordance with the graphic standards included in MIL-STD-1840.

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3.3.3.4 Audio, video, and process. The audio, video, and process elements shall provide the capability for the author to define an audio sequence, a video sequence, or a call to a software process.

3.3.3.5 Dialogs. Dialog elements are the basic element which provides the capability for user interaction with the TI. During a presentation these elements shall be used to prompt the user to input a response ("fillin"), select a choice from a set of alternatives ("menu"), or to select items from within a text, table or graphic ("selection").

3.3.4 Context dependent filtering. Context dependent filtering shall be accomplished through author-defined preconditions. Preconditions shall contain an expression which will contain all the information necessary to identify what conditions must be present to display the TI.

3.3.4.1 Preconditions. A precondition shall contain an expression which identifies the conditions which must be present to display the TI. Precondition elements may be referenced by node elements. This implies that the element's information is relevant only if the precondition is true in the presentation situation.

3.3.4.2 Postconditions. Postconditions shall assert the value of an expression to a property. Once these property values are asserted, they shall be accessible to the presentation software for later testing and processing to determine the user's situation.

3.3.4.3 Expressions. Expressions developed for an IETMDB shall conform to one of four types of expressions defined in the CDM. The first is a binary operation between two expressions; the second is a unary operation which is applied to an expression; the third and fourth are operations that identify a unique property (variable) or a value to be used in an expression.

3.4 Content specific layer. All TI shall be structured in accordance with a content specific DTD. One content specific DTD shall apply for an entire set of information regardless of the desired access to the information. The CDM shall define the content and structure of the TI but shall not describe format information.

3.4.1 Control of content specific DTDs. The contractor shall not exchange TI with the DoD unless it has been developed in accordance with the generic layer DTD and one or more of the latest versions of DoD approved content specific DTDs. If a content specific DTD does not exist which meets the contract's requirements, the contractor shall submit a content specific DTD to the Government for approval.

3.4.2 Development of content specific DTDs. If a new content specific DTD is developed, the contractor shall ensure that the content specific DTD meets the requirements of ISO 8879, and the requirements imposed by the generic layer DTD.

3.4.2.1 Use of generic DTD primitive elements. The generic layer of the CDM shall define a set of primitive elements. Those elements shall be available to any content specific layer DTD that includes the generic layer in an entity declaration and corresponding entity reference. Any element defined within a content specific DTD which requires the use of any of the primitive elements need only include text, table, graphic, or dialog within its content model. The contractor shall not redefine primitive elements within the content specific DTD. Those elements, using primitive

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elements, shall be restricted to the structure of primitive elements as defined within the generic layer.

3.4.2.2 Use of generic DTD template elements. Elements within a content specific DTD shall conform to one of the templates defined within the generic layer. Elements shall include the attributes listed under the generic layer's definition of the templates. The two common attributes among the five templates are identification (id) and CDM. Each element employing a template includes an identification attribute for referencing. The CDM attribute identifies which template an element is employing.

3.4.3 Content specific DTD for Organizational Level (O-Level) maintenance. The following describes requirements for the content specific DTD included in this specification:

3.4.3.1 Item/System hierarchy. The vehicle, weapon system, or other equipment that is being maintained and operated is composed of several layers of subsystems, components, and parts. This hierarchical representation shall be accomplished by use of a system element that is used recursively, and which breaks down the equipment into only those components that are being maintained or operated. Each component of this hierarchy shall have one or more of the following four categories of information associated with it:

a. Descriptive information

b. Procedural information

c. Fault isolation information

d. Parts information

3.4.3.2 Descriptive information. Descriptive information shall contain a hierarchy of narrative paragraphs. Paragraphs, in turn, may refer to primitive elements. Descriptive information may provide information on system (subsystem, component, part) physical arrangement, functional behavior, theory of operation, and other aspects.

3.4.3.3 Procedural information. Procedural information shall be composed primarily of task statements. Each task element shall be associated with attributes which provide related information such as: estimated completion time; maintenance level(s) where the task is to be performed; required conditions which must be met before performing the task; and the number of people required to perform the task. A procedural element may be linked to other elements which define the support equipment and consumables that task requires, through the establishment of appropriate relationships.

3.4.3.4 Fault isolation information. Fault isolation information shall contain data necessary to isolate faults found in a system. Fault isolation information shall contain fault elements, fault state elements, test elements, outcome elements, and rectification elements.

3.4.3.4.1 Fault elements. Fault elements shall identify potential faults which might occur in the system.

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3.4.3.4.2 Fault state elements. Fault state elements shall present a list of faults implicated as the result of a test that has been performed. Each suspected fault in the list shall be weighted, based on the probability that it is the cause of the observed malfunction. The fault state element may also present a list of possible faults that have been eliminated from consideration as the result of tests performed.

3.4.3.4.3 Test elements. Test elements shall contain a link to the procedural instructions a technician must follow to carry out a required task at a particular juncture in the fault isolation procedure. Test elements shall also provide all possible test outcomes.

3.4.3.4.4 Outcome elements. Outcome elements shall contain definitions of new fault states associated with the results of a particular test. Outcome elements shall also contain a description of the state of the item being maintained. An outcome is based on one or more expressions (i.e., system states which must be established for the specific outcome to apply). The final outcome element of a fault isolation procedure shall have a relationship which associates it with an identified fault. The identified fault has, in turn, associated with it the initial element of the appropriate corrective maintenance action.

3.4.3.4.5 Rectification elements. Rectification (i.e., corrective maintenance actions) elements shall contain references to procedural rectification tasks, checkout tests used to report the success of completed rectification tasks, and a list of all faults that the rectification shall repair.

3.4.3.5 Parts information. Two types of parts information shall be included: (1) maintainer/operator information, and (2) supply information. Elements containing either type shall refer explicitly to corresponding elements of the other type.

3.4.3.5.1 Parts information for the maintainer or operator. Parts information provided for a system maintainer or operator shall include such items as units per assembly, usable-on code, Mean Time Between Failures (MTBF), and reference designator, if applicable.

3.4.3.5.2 Parts information provided for parts supply. Parts information provided for the parts supply process shall constitute unambiguous identification of a part so that it can be reordered, and may consist of such items as: the part number; Commercial and Government Entity (CAGE) code; Source, Maintenance, and Recoverability (SMR) code; Hardness Critical Item (HCI) identification; and National Stock Number (NSN), if applicable.

4. VERIFICATION.

4.1 Verification. Unless otherwise specified in the contract or purchase order:

a. Validity of the accuracy and scope of the IETMDB technical content, user interface functionality, and EDS-IETM interface shall be the responsibility of the contractor (see 6.2).

b. The contractor shall provide suitable facilities to perform the validation functions specified herein.

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c. The contractor's existing quality assurance procedures shall be used.

d. The government reserves the right to review any of the verifications.

4.1.1. Minimum verification requirements. As a minimum, verification shall ensure the following:

a. Suitability of the IETMDB for the intended maintenance environment.

b. Usability by the intended users.

c. Compatibility with other Government systems.

4.1.2 Compliance. All IETMDB shall meet all of the requirements of sections 3 and 5 of this specification. The requirements set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any requirements in this specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Use of sampling inspections shall be in accordance with commercially acceptable quality assurance procedures; however, Government approval for use of sampling in QA procedures does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

5. PACKAGING.

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. An IETMDB is the source data for the preparation of IETMs. IETMs prepared in accordance with this specification are intended for use in the installation, operation, maintenance, repair, and logistics support of equipment/systems or for the accomplishment of the assigned mission of users.

6.1.1 Nature and purpose of a revisable source data base. For complex weapon systems and other types of military equipment, adequate logistic support in all its forms requires an enormous amount of current, readily accessible, accurate, and highly detailed data, consisting of TI. This information has been traditionally prepared and distributed to the end user in paper form; but with new technology, it can be better and more effectively displayed or presented electronically and interactively to an end user. The material presented is derived from material stored in textual, graphical, audio, or video form in a revisable data base which is composed of logically connected but randomly accessible IETM data elements. It is this starting point of the IETM electronic data chain that is specified in this document. An integral part of the IETM concept and, in the larger arena of the Department of Defense (DoD) Computer-aided Acquisition and Logistic Support (CALS) program, is that the Services can acquire and maintain large scale data bases. They can also gain access to such data bases that are maintained continuously by a contractor.

6.1.2 IETMDB capabilities. An IETMDB is a complete collection of data base elements relating to a weapon system or other equipment acquired by the Government and constructed in a standardized manner to provide the following capabilities:

a. The IETMDB can serve as the basis for construction and update of the entire suite of electronically-displayed, weapon system, IETMs through the use of automated authoring systems.

b. Government activities or DoD contractors concerned with logistic support for the weapon system involved, can access the data base directly to obtain needed logistic support information for specific purposes.

c. The IETMDB, or portions of it, can be interchanged by means of standardized formats and procedures throughout the DoD and its supporting contractors when needed for any purpose.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of the DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1, 2.2.2).

c. If IETM program requires content specific layer elements other than those already specified in Appendices B and D (see 3.1).

d. If the specification applies to the delivery and tagging of an IETMDB (see 3.1.1).

e. If the specification applies to the structuring of the IETMDB and naming of the IETMDB elements which are created and maintained by the contractor (see 3.1.2).

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f. Content specific DTDs other than the one included herein and whether these are to be used in addition to or instead of the content specific DTD included herein (see 3.2.4).

6.2.1 Technical information procurement options. Acquisition of IETMs may be carried out by one of several optional approaches. This specification provides requirements for a standardized IETMDB which will permit the Government to acquire TI by applying any of the following contractual options:

a. Acquisition of only the final form IETMs which are required. Although the author (equipment prime contractor) will need to establish an automated equipment or weapon-system (source) data base, this data base will not be acquired by the Government. The contractor will maintain, use, and control the data base, both for the preparation of IETMs and for other purposes. The Government under this specification requires that the data base be structured and the individual data elements named and attributed in a standard manner. However, an explicitly tagged data file need not be prepared for delivery as no data base delivery is required.

b. Acquisition of the IETMDB. Acquisition of the IETMDB may involve either of the following options:

(1) Delivery to the Government, in standardized form, and subsequently maintained by the Government (with or without update information supplied on a continuing basis by the contractor).

(2) Title acquired to the IETMDB by the Government, but with the data base retained and maintained in standardized form in the contractor's plant. The Government could be provided with on-line access to the data base.

c. Acquisition of fully constructed IETMs (fully prepared and validated by the contractor), as well as the IETMDB upon which they are based. Acquisition under this option may involve either option (1) or (2) as given in 6.2.1b above.

6.3 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on contract. If technical manuals are required, performance specifications, and standards that have been listed in DoD 5010.12L, Acquisition Management System and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included to the contract. The technical manuals must be acquired under a separate line item in the contract.

6.4 Definitions of acronyms and terms. Acronyms and IETM terms not listed in MIL-STD-12 are included in the definitions contained in 6.4.1 through 6.4.2.

6.4.1 Acronyms.

ANSI	American National Standards Institute
BL	Buttock Line
CAGE	Commercial And Government Entity
CALS	Computer-aided Acquisition and Logistics Support

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CDM	Content Data Model
DBMS	Data Base Management System
DD	Department of Defense (document-number prefix)
DoD	Department of Defense
DoDISS	Department of Defense Index of Specifications and Standards
DTD	Document Type Definition
EDS	Electronic Display System
FS	Fuselage Station
HCI	Hardness Critical Item
ICC	Item Category Code
IEC	International Electrotechnical Commission
IETM	Interactive Electronic Technical Manual
IETMDB	IETM Data Base
ISO	International Organization for Standardization
LRU	Line Replaceable Unit
MTBF	Mean Time Between Failures
NSN	National Stock Number
QA	Quality Assurance
SGML	Standard Generalized Markup Language
SMR	Source, Maintenance, and Recoverability (Code)
STD	Standard
TI	Technical Information
WL	Water Line
WS	Wing Station

6.4.2 Interactive Electronic Technical Manual (IETM). An IETM is a technical manual, prepared (authored) by a contractor and delivered to the Government, or prepared by a Government activity, in digital form. The IETM is developed using a suitable authoring tool that possess the following characteristics:

a. The format and style of the presented information are optimized for screen presentation to assure maximum comprehension; that is, the presentation format is "information oriented", not "page oriented".

b. The elements of technical data that makes up the IETM is so interrelated that a user's access is made as easy as possible, and is achieved through a variety of paths.

c. The computer controlled IETM display device can function interactively (as a result of user requests and information input) in providing procedural guidance, navigational directions, and supplemental information. It also provides assistance to carry out logistic support functions, supplemental to maintenance.

6.5 Definitions.

6.5.1 Verification. Verification (section 4), in the context of this specification equates to the contractor's quality assurance program for validating the content of the IETM. Suggested validation methods include:

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a. Actual performance. Using production configured equipment, hands-on performance of the procedure using the technical instructions as written.

b. Simulation. Using production configured equipment and the technical manual procedure, simulate the actions required by comparing the task steps to the hardware, while not actually removing any equipment.

c. Table top analysis. Primarily for non-procedural data, compares the technical content to source data to ensure the technical accuracy and depth of coverage.

6.5.2 Subject terms (key word) list.

Database
Interactive Electronic Technical Manual (IETM)
Interactive Electronic Technical Manual Data Base (IETMDB)
Content Data Model (CDM)
Technical Manuals

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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APPENDIX A

GENERIC LAYER
DOCUMENT TYPE DEFINITION (DTD)

A.1 SCOPE.

A.1.1 Scope. The DTD within this appendix provides the structure and content of documents prepared in accordance with this specification. Unless otherwise specified by the procuring activity, this Appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS.

A.2.1 Government documents.

A.2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

ISO 8879	Information Processing - Text and Office Systems - Standard Generalized Markup Language (SGML)
ISO 10744	Information Technology - Hypermedia/Time-Based Document Structuring Language (HyTime)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

A.3 GENERIC LAYER DOCUMENT TYPE DEFINITION.

A.3.1 Use of SGML. The markup tags described herein are based on rules outlined in ISO 8879. All data to be delivered digitally in accordance with this specification shall be tagged using the SGML declaration in section A.3.1.1 of this document, the DTD in this section, and associated content specific DTD(s).

A.3.1.1 SGML declaration. The SGML declaration for this specification is as follows:

```
<!SGML "ISO 8879:1986"
CHARSET     -- ASCII character set --
            BASESET     "ISO 646-1991//CHARSET International
            Reference Version (IRV)//ESC 2/5 4/0"
            DESCSET     0     9     UNUSED
                        9     2     9           -- TAB, LF --
                        11    2     UNUSED
                        13    1     13          -- CR --
                        14    18    UNUSED
                        32    95    32
                        127   1     UNUSED
            -- Additional character set per MIL-PRF-28001B --
```


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```
BASESET     "ISO Registration Number 100//CHARSET ECMA-94
               Right Part of Latin Alphabet Nr. 1//ESC 2/13
                 4/1"
            DESCSET     128   32    UNUSED
                        160   5     32
                        165   1     UNUSED
                        166   88    38
                        254   1     127
                        255   1     UNUSED
CAPACITY    SGMLREF     TOTALCAP          32165152
                        ENTCAP            3000000
                        ENTCHCAP          3000000
                        ELEMCAP           3000000
                        GRPCAP            3000000
                        EXGRPCAP          3000000
                        EXNMCAP           3000000
                        ATTCAP            3000000
                        AVGRPCAP          3000000
                        IDCAP             3000000
                        IDREFCAP          3000000
SCOPE       DOCUMENT
SYNTAX      SHUNCHAR    CONTROLS    0 1 2 3 4 5 6 7 8 9 10 11 12 13 14
                                    15 16 17 18 19 20 21 22 23 24 25 26
                                    27 28 29 30 31 127 255
            BASESET     "ISO 646-1991//CHARSET International
Reference Version (IRV)//ESC 2/5 4/0"
            DESCSET     0   128   0
            FUNCTION    RE                13    -- CR --
                        RS                10    -- LF --
                        SPACE             32    -- SP --
                        TAB SEPCHAR       9     -- TAB --
            NAMING      LCNMSTRT          ""    -- in addition to a..z --
                        UCNMSTRT          ""    -- in addition to A..Z --
                        LCNMCHAR          "-."  -- - , .in addition to
a..z, 0..9 --
                        UCNMCHAR          "-."  -- - , . in addition to
A..Z, 0..9 --
                        NAMECASE    GENERAL     YES
                                    ENTITY      NO
            DELIM       GENERAL     SGMLREF
            SHORTREF    NONE
            NAMES       SGMLREF
            QUANTITY    SGMLREF     ATTCNT      400
                        ATTSPLEN    30000
                        ENTLVL      1600
                        GRPCNT      253
                                    GRPGTCNT    253
                                    GRPLVL      253
                                    LITLEN      30000
                                    NAMELEN     32
                                    TAGLEN      30000
                                    TAGLVL      240
FEATURES    MINIMIZE
                  DATATAG     NO
                  OMITTAG     YES
                  RANK        NO
                  SHORTTAG    NO
            LINK
```


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```
                SIMPLE      NO
                IMPLICIT    NO
                EXPLICIT    NO
          OTHER
                CONCUR      NO
                SUBDOC      NO
                FORMAL      YES
APPINFO     "HYTIME"
>

<?HyTime VERSION "ISO/IEC 10744:1992">
<?HyTime MODULE base refctl exidrefs>
<?HyTime MODULE locls multloc anysgml anydtd mixspace>
<?HyTime MODULE links manyanch>
```

A.3.2 Template document type. The DTD fragment for this specification is as follows:

```
<!-- ******************************************************

              IETM CONTENT DATA MODEL    Version 6.1
              Generic Layer           1 October 1992

     ******************************************************
```

The IETM CDM provides a representation of technical information elements and their relationships. The CDM is composed of two separate layers. The first is the "Generic Layer". It defines general characteristics which are common across all applications. The second layer is the "Content Specific Layer," which contains content specific DTDs.

The generic layer defines the templates, linking elements, primitive elements, and context filtering elements which are used to create content specific DTDs. Templates define rules which must be followed in the creation of content specific DTD's and document instances. The templates provide the structure for creating composite nodes, context dependent filtering, user interaction and branching. The templates provide basic sets of rules to which elements must adhere. Those rules are explained after the declaration of each template in this document.

The CDM hylink element is taken from the HyTime Model. This element provides the capability to link between CDM elements, other SGML files, and non SGML documents. These capabilities are explained in the HyTime Linking Mechanism section of this document. The CDM "link" element is a non-HyTime linking mechanism. It provides the capability to link between CDM elements only.

The primitive elements ("text", "table", "graphic", "audio", "video","process", and "dialog") are defined in the generic layer. These elements may be used to construct a variety of composite elements in the content specific layer. The primitive elements' structures shall remain constant.

Context dependent filtering provides the capability to present the user with only the information that applies to a specific situation. The precondition and postcondition elements provide the mechanism for context dependent filtering. The precondition element enables the selection of the appropriate information for presentation. The postcondition element enables the recording of presentation events for later filtering.

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```
     ********************************************************

                    PUBLIC ENTITY DECLARATIONS

     ********************************************************

<!ENTITY % dietmdb-a PUBLIC "-//USA-DOD//DTD Content Data Model Generic
Layer//EN">  -->

<!-- Inclusion of MIL-PRF-28001 math package -->

<!ENTITY % mathpac PUBLIC "-//USA-DOD//DTD SUP MIL-PRF-28001 MATHPACK
911001//EN" >
%mathpac;
<!ENTITY % mathtxt "dfref | f" >
<!ENTITY % mathcon "df | dfg" >

<!--  The following entity declarations provide a mechanism for referencing
primitive elements in the generic layer, and within any content specific DTD.
In SGML, an entity must be declared prior to an element referencing that
entity.  Therefore, all primitive entities have been moved to this section.  A
detailed description of each primitive will appear when the element is
declared later in this document.  -->

<!ENTITY % text    "text | text-alts" >
<!ENTITY % table   "table | table-alts" >
<!ENTITY % graphic "graphic | graphic-alts | grphprim | grphprim-alts" >
<!ENTITY % audio   "audio | audio-alts" >
<!ENTITY % video   "video | video-alts" >
<!ENTITY % process "process | process-alts" >
<!ENTITY % dialog  "dialog | dialog-alts" >
<!ENTITY % link    "link | hylink" >

<!--  The following entity provides a simple method for referencing the
primitive elements defined in the generic layer.  -->

<!ENTITY % primitive  " %text; | %table; | %graphic; | %audio; | %video; |
         %process; | %dialog; | expression | assertion " >

<!ENTITY % linkendlist "(descinfo | partinfo | text | table | graphic | audio
         | video | para | task | partbase | process | dialog | expression |
         assertion | entry )">

<!-- ********************************************************

                      NOTATION DECLARATIONS

     ********************************************************

   The following notations define external references to  "public" graphics
standards used in the CDM.  The specified abbreviations (cgmbin, cgmclear,
cgmchar, fax, faxtile, iges) are used by the element "grphprim" to specify the
type of graphic representation used to encode a particular graphic primitive.
-->

<!NOTATION   cgmbin      PUBLIC "ISO 8632/2//NOTATION Binary encoding//EN">

<!NOTATION   cgmchar     PUBLIC "ISO 8632/2//NOTATION Character encoding//EN">
```


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```
<!NOTATION   cgmclear    PUBLIC "ISO 8632/2//NOTATION Clear text
encoding//EN">

<!NOTATION   fax         PUBLIC "-//USA-DOD//NOTATION CCITT Group 4 Facsimile
Type 1 Untiled Raster//EN">

<!NOTATION   faxtile     PUBLIC "-//USA-DOD//NOTATION CCITT Group 4 Facsimile
Type 2 Tiled Raster//EN">

<!NOTATION   iges        PUBLIC "-//USA-DOD//NOTATION (ASME/ANSI Y14.26M-
1987)Initial Graphics Exchange Specification//EN">


<!-- ******************************************************

                              ROOT ENTITY

     ******************************************************

   The a.root entity enables a content specific layer to comply with the
support requirements for HyTime.  This entity is only used in the root element
of the DTD.  -->

<!ENTITY % a.root
                    "HyTime    NAME       HyDoc
                     boslevel  NUMBER     #IMPLIED" >


<!-- ******************************************************

                            TEMPLATES

     ******************************************************

   The following section defines the generic layer templates.  These templates
define semantic rules for creating content specific elements.  These semantic
rules make up the minimum set of constraints on content specific elements.

   There are two general rules to follow when creating a content specific
element.  First, the element's content model must comply with the template's
content model.  Second, the template's attribute entity must be included in
the element's attribute list.  The attribute entities for all templates
include the attributes "id", "cdm", "ref".  The "cdm" attribute indicates
which template the element is employing.  The "id" and "ref" attributes are
used for non-redundant referencing and linking.

   The "ref" attribute utilizes the SGML #CONREF capability.  A #CONREF
attribute is only filled in when the element's content model is empty.  In
this case, the #CONREF attribute contains a reference which is a unique
identifier to either an element of the appropriate type or a location element
that resolves to an element of the appropriate type (see section on Hytime
linking mechanism).  When an element uses the #CONREF capability, the
referencer's attribute list will take precedence over the referenced element's
attributes.

   The "hylink" element utilizes the HyTime link capability.  In this case, a
hylink is a reference which is a unique identifier to a location element that
```


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resolves to an element of the appropriate type (see section on HyTime linking mechanism).

The "link" element is a simpler version of the HyTime linking mechanism. This link provides a capability to link only within the IETMDB using the SGML #CONREF feature.

This section includes an element declaration for each template (NODE, NODE ALTS, NODE SEQ, IF NODE, LOOP NODE). The declarations are enclosed within comments, and are not formally a part of the DTD. These element declarations use template names, in all caps, to describe content model constraints for each template. When creating content specific elements, these template names must be replaced by element names of the appropriate type. -->

```
                    <!-- *****  NODE TEMPLATE  ***** -->

<!-- The NODE contains the content of the technical information.  The NODE
element creates hierarchy within the CDM.  NODE also contains context
filtering preconditions and postconditions.  The link entity within the NODE
provides the capability to cross reference other technical information, both
internal (link element) and external (hylink element) to the IETMDB.  The use
of link, from the Hytime model, provides additional functionality by allowing
a link to be made to a document outside the CDM specification boundary.

   The NODE template provides the capability to create composite structures
within the content specific layer. Composite structures may contain
subcomponents that employ the NODE, NODE ALTS, or NODE SEQ templates.  The
NODE subcomponents may be composite structures themselves or they may be
primitive NODES (text, tables, graphics, audio, video, process, and dialog).
Composite structures create hierarchy within the CDM.  When composite nodes
contain other composite nodes there is an implied hierarchy.  The composite
node in the content model is at a lower level in the hierarchy (e.g. a Task
Node contains a  Step-Seq Node in its content model, which in turn contains
Step Nodes).

      The following defines the NODE template:

<!ELEMENT "NODE"  - -  ( precond*, (%link;)*, ( NODE | NODE-ALTS | NODE-SEQ |
                       %primitive; )*, postcond* )>
-->

<!ENTITY % a.node
                     "id        ID         #IMPLIED
                      name      CDATA      #IMPLIED
                      type      CDATA      #IMPLIED
                      itemid    CDATA      #IMPLIED
                      cdm       NAME       #FIXED     'node'
                      ref       IDREF      #CONREF" >

<!--  The following semantic rules apply to any content specific element
employing the NODE template: (1) The element may contain a list of
preconditions that identify the element's applicability.  The list of
preconditions will be evaluated at presentation time, and if all preconditions
evaluate to true, that node will be presented. (2) The element may contain
relational links to other data items.  (3) The element may contain
subcomponents that employ the NODE, NODE ALTS, or NODE SEQ templates.  (4) The
element may contain a list of postconditions which record presentation events.
The  postconditions will be evaluated after the NODE and all its subcomponents
```


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```
have been presented.  The postcondition values will then be assigned to their
specified properties.  -->

                    <!-- *****  NODE ALTS TEMPLATE  ***** -->

<!--  NODE ALTS  (node alternatives) will contain a list of mutually exclusive
nodes.  Their grouping is due to the fact that they apply in different
contextual situations.  In this manner, the NODE-ALTS element is a logical
reference that contains a set of NODES which might apply to different
situations.  An important fact in the NODE-ALTS  structure is that no
hierarchy is implied between the generic identifier and the content model
NODES (e.g. a Task-alts element will contain Task-nodes in its content model).

      The following defines the NODE ALTS template:

<!ELEMENT "NODE-ALTS"       - -        ( NODE )+ >

-->

<!ENTITY % a.node-alts
          "id       ID        #IMPLIED
           cdm      NAME      #FIXED    'node-alts'
           ref      IDREF     #CONREF"                  >

<!--  The following semantic rules apply to any content specific element
employing the NODE ALTS template. (1) The element must contain components that
employ the NODE template.  (2) The components must be of the same element type
and at the same level in the hierarchy.  (3)   At presentation time, the
precondition for each NODE alternative will be evaluated.  The NODE whose
precondition evaluates to "true" will be selected for presentation.  (4) These
components must contain mutually exclusive preconditions.  In any specific
situation, at most one node would have a precondition which evaluates to true.
(5) There need not be an applicable component for every possible situation.
-->
                    <!-- *****  NODE SEQ TEMPLATE  ***** -->

<!--  The NODE SEQ template is the mechanism for creating interactive
sequences with the user.

     The following defines the NODE SEQ template:

<!ELEMENT "NODE-SEQ"   - -    ( NODE | NODE-ALTS | IF-NODE | LOOP-NODE )+ >

-->

<!ENTITY % a.node-seq
          "id       ID        #IMPLIED
           cdm      NAME      #FIXED    'node-seq'
           ref      IDREF     #CONREF"                  >

<!--  The following semantic rules apply to the NODE SEQ template.  (1) Any
content specific element employing NODE-SEQ must contain components that
employ the NODE, NODE ALTS, IF NODE, or LOOP NODE templates.  (2) The
components of a NODE SEQ are always traversed in the order they appear.  This
traversal includes the branching and iteration implicit in any IF NODES or
LOOP NODES in the sequence logic.  -->

                    <!-- *****  IF NODE TEMPLATE  ***** -->
```


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```
<!--  The IF NODE template uses the same logic as the IF-THEN-ELSE statement
in a programming language.  The "IF" part is the expression in the content
model.   The "THEN" part is the first NODE SEQ; the "ELSE" part is the second
NODE SEQ, which is optional.

     The following defines the IF NODE template:

<!ELEMENT "IF-NODE"   - -     ( expression, NODE-SEQ, NODE-SEQ? ) >

-->

<!ENTITY % a.if-node
          "id       ID        #IMPLIED
           cdm      NAME      #FIXED    'if-node'
           ref      IDREF     #CONREF"                  >

<!--  The following semantic rules apply to the IF NODE template.  (1)  The
expression will be evaluated at presentation time; (2) If the expression
evaluates to "true" the first NODE SEQ will be traversed; (3) If the
expression evaluates to anything but "true", and the second NODE SEQ is
present, the second NODE SEQ is traversed. (4) If the expression evaluates to
anything but "true", and the second NODE SEQ is not present, the next element
in the sequence will be presented. -->

                    <!-- *****  LOOP NODE TEMPLATE  ***** -->

<!--  The LOOP NODE template provides the capability similar to that found in
a programming language for creating loops.  The template provides the syntax
for creating WHILE or FOR NEXT loops, whichever applies to the situation.  For
example, when creating a FOR NEXT loop, the first assertion initializes the
control variable for the loop.  The expression is the test criterium for
exiting the loop.  The second assertion alters the control variable at the end
of each loop iteration.  The node sequence provides the actual element(s) to
be repeated within the loop.

     The following defines the LOOP NODE template:

<!ELEMENT "LOOP-NODE"    - -    ( assertion?, expression, assertion?,
                                  NODE-SEQ) >
-->

<!ENTITY % a.loop-node
          "id       ID        #IMPLIED
           cdm      NAME      #FIXED    'loop-node'
           ref      IDREF     #CONREF"                      >

<!--  The following semantic rules apply to the LOOP NODE template, when
employing it as in a FOR NEXT loop.  (1) At the beginning of the loop the
first assertion is evaluated and the value is assigned to the specified
property.  (2) The expression is evaluated and if the expression evaluates to
anything but "true" the loop is terminated.  (3) If the expression evaluates
to true, the NODE SEQ is traversed. (4)  At the end of each iteration, the
second assertion is evaluated and the value is assigned to the specified
property. (5) Steps 2-4 are continued until the loop terminates.

   The semantic rules which apply to the LOOP NODE template, when employing it
as in a WHILE loop, are as follows.  (1)  The expression is evaluated and if
```


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```
the expression evaluates to anything but "true" the loop is terminated.  (2)
If the expression evaluates to true, the NODE SEQ is traversed.  (3)  Steps 1-
2 are continued until the loop terminates.

     ******************************************************

                    LINKING MECHANISM

     ******************************************************

   This section defines the simple linking mechanism used for linking internal
to the IETMDB.  -->

<!ELEMENT link     - -      ( #PCDATA )  >
<!ATTLIST link
          id             ID             #IMPLIED
          endtypes       %linkendlist;  #REQUIRED
          linkends       IDREFS         #REQUIRED  >

<!--  The "link" element provides the capability for creating relational links
within the CDM.  "Link" is included within the content model of the NODE
template, therefore, any content specific element employing the NODE template
may include relational links.

  The 'endtypes' attribute identifies the type of primitive or element that
the link is pointing to, and the 'linkends' attribute contains the unique
identifier attribute (id) of the element being pointed to.

     ******************************************************

                    HYTIME LINKING MECHANISM

     ******************************************************

     This section defines the linking mechanism based on the HyTime standard.
-->

<!ELEMENT hylink     - -      ( #PCDATA )  >
<!ATTLIST hylink
          HyTime    NAME      #FIXED 'ilink'
          id        ID        #IMPLIED
          anchrole  NAMES     #FIXED 'hotspot target'
          linkends  IDREFS    #REQUIRED
          reftype   CDATA     #FIXED 'linkends linkendtypes #SEQ'
          extra     NAMES     'A A'
          intra     NAMES     'A A'
          endterms  IDREFS    #IMPLIED
          aggtrav   NAMES     agg>

<!--  The "hylink" element provides the capability for creating HyTime-
compliant relational links within the CDM.  "Hylink" is included within the
content model of the NODE template, therefore, any content specific element
employing the NODE template may include relational HyTime links.  The
'anchrole' attribute identifies the type of primitive or element that the link
is pointing to, and the 'linkends' attribute contains the unique identifier
attribute (id) of the element being pointed to. -->
```


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```
<!ELEMENT linkendtypes   - - (%link;) (link | descinfo | partinfo | text |
                                table | graphic | audio | video | process
                                |para |task | partbase | dialog |entry)*)>

<!ELEMENT nmlist     - -   (#PCDATA)>
<!ATTLIST nmlist
          HyTime    NAME           nmlist
          nametype  (entity|element|unified)  #REQUIRED
          obnames   (obnames|nobnames) nobnames
          docorsub  ENTITY         #IMPLIED
          dtdorlpd  NAMES          #IMPLIED >

<!ELEMENT nameloc   - - (nmlist)* >
<!ATTLIST nameloc
          HyTime    NAME           nameloc
          id        ID             #REQUIRED
          ordering  (ordered|noorder) ordered
          set       (set|notset)   notset
          aggloc    (aggloc|agglink|nagg) agglink >

<!-- ******************************************************

                    PRIMITIVE ELEMENT DECLARATIONS

     ******************************************************

   The following element declarations define the primitive data elements used
throughout the technical information.  Each element is defined in terms which
can be employed in any content specific DTD.   -->

                          <!-- *****  TEXT  ***** -->

<!ELEMENT textcont       - o     ( precond*, link, ( textcont |
                                   text-alts | text | %mathtxt;
                                   | %mathcon; ) + ) >

<!ELEMENT text                     - -     ( #PCDATA ) >
<!ATTLIST text

<!--  A "text" unit is basically a text string of "parsable character data" or
PCDATA.  Within a text string, there may be embedded "text" elements which
allow the referencing of other elements or parts of elements through the link
or hylink/location mechanism explained in the HyTime section of this document.
Those embedded  "text" references are inserted in the text string that
contained them.  For example, the string may contain a reference to a standard
system name, a standard part nomenclature or a standard task name.  By using
this mechanism, standard terminology can be referenced consistently throughout
the data base, and any changes to the standard terminology can be made in one
location and automatically updated throughout the data base.  -->

<!ELEMENT text-alts      - -     ( text )+ >
<!ATTLIST text-alts
          %a.node-alts;                    >

<!--  This element provides the ability to have context sensitive filtering of
text.  -->

                          <!-- *****  TABLE  ***** -->
```


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```
<!ELEMENT table       - -      ( precond*, (%link;)*, rowhddef*,(colhddef?,
                                 entry+ )+ ) >
<!ATTLIST table
          %a.node;                              >

<!--  This element defines how a table is constructed.  A "table" will contain
a list of the row headers.  Each column header will be followed by one or more
entries.  The combination of column header and entries may be repeated for as
many columns as the "table" requires.  -->

<!ELEMENT table-alts      - -        ( table )+ >
<!ATTLIST table-alts
          %a.node-alts;  >

<!--  This element provides the ability to have context sensitive filtering of
"tables".  -->

<!ELEMENT rowhddef    - -        ( %text; ) >
<!ATTLIST rowhddef
            id              ID          #IMPLIED
            ref             IDREF       #CONREF
            row             NUTOKEN     #REQUIRED >

<!--  This element defines a row header to be a piece of text and a row
number.  However, if a row header has been previously defined, the 'ref'
attribute allows the referencing of that element from another table.  The
'row' for the element that references a previously defined header takes
precedence over the original 'row' in the referenced header.  -->

<!ELEMENT colhddef    - -        ( %text; ) >
<!ATTLIST colhddef
            id              ID          #IMPLIED
            ref             IDREF       #CONREF
            colnum          NUTOKEN     #REQUIRED >

<!--  This element defines a column header to be a piece of text and a column
number.  However, if a column header has been previously defined, the 'ref'
attribute allows the referencing of that element from another table.  The
'colnum' for the element that references a previously defined header takes
precedence over the original 'colnum' in the referenced header.  -->

<!ELEMENT entry       - -        ((%link;)*, (%text; | %graphic; )) >
<!ATTLIST entry
            id              ID          #IMPLIED
            ref             IDREF       #CONREF
            colnum          NUTOKEN     #REQUIRED
            row             NUTOKEN     #REQUIRED >

<!--  This element defines an entry for a cell in a table to be a piece of
text and the appropriate row and column.  The row and column define the cell
to place the entry.  However, if an entry has been previously defined, the
'ref' attribute allows the referencing of that element from another table.
The 'colnum' and 'row' for the element that references a previously defined
entry take precedence over the original 'colnum' and 'row'. -->

                    <!-- *****  GRAPHICS  ***** -->
```


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```
<!--  The CDM allows the referencing of "graphics" in external representations
or embedded within the CDM.  Graphics are an integral part of technical
information.  Therefore, all possible standard representations have been
included within the following primitive elements.  -->

<!ELEMENT graphic    - -     ( precond*, (%link;)*, ( %graphic;)+ ) >
<!ATTLIST graphic
     %a.node;
     minsize        NUTOKENS            #IMPLIED
     penshape       CDATA               #IMPLIED
     penpatt        CDATA               #IMPLIED
     transfrm       NUTOKENS            #IMPLIED
     window         NUTOKENS            #IMPLIED >

<!--  This element describes graphics in terms of primitives or references to
other graphics, thus providing the ability to create composite graphics.  -->

<!ELEMENT graphic-alts    - -       ( graphic )+ >
<!ATTLIST graphic-alts
          %a.node-alts;                          >

<!--  This element provides the ability to have context sensitive filtering of
graphics.  -->

<!ELEMENT grphprim     - -    ( precond*, (%link;)*, (%text;) ) >
<!ATTLIST grphprim
     %a.node;
     coding         NOTATION  (cgmchar _  cgmbin  | cgmclear _
                               fax     _  faxtile _ iges    )     'cgmbin'
     minsize        NUTOKENS            #IMPLIED
     penpatt        CDATA               #IMPLIED
     penshape       CDATA               #IMPLIED
     transfrm       NUTOKENS            #IMPLIED
     x-location     NUTOKEN             #IMPLIED
     y-location     NUTOKEN             #IMPLIED
     window         NUTOKENS            #IMPLIED
     external-ptr   ENTITY              #IMPLIED
     picid          NUTOKEN             #IMPLIED        >

<!--  This element defines a primitive graphic which may be contained in the
content or referenced by the 'ref' attribute. The graphic is represented in
one of the valid formats (cgmbin, cdmchar, cgmclear, fax, faxtile, iges), and
the format is indicated by the coding attribute.  The 'type' attribute may
identify a graphic as a "hotspot", thus making it selectable during
presentation.  The minsize attribute specifies the minimum height requirements
for display of the graphic.  Any transformations or manipulations of the
graphic, other than those described by the notations, can be defined using the
penpatt, penshape, transfrm, or window attributes on the graphic
primitive element.  -->

<!ELEMENT grphprim-alts        - -      ( grphprim )+ >
<!ATTLIST grphprim-alts
          %a.node-alts;                              >

<!--  This element provides the ability to have context sensitive filtering of
graphic primitives.  -->

                <!-- *****  AUDIO, VIDEO & PROCESS  ***** -->
```


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```
<!-- The elements "audio", "video", "process" provide the capability for the
author to define an audio sequence, video sequence, or a call to a software
process.  These element definitions require further inspection and updating,
which will be done upon completion of a closer look at the HyTime multi-media
event definitions. -->

<!ELEMENT audio     - -      ( precond*, (%link;)* ) >
<!ATTLIST audio
          %a.node;
          external-ptr   IDREF   #REQUIRED  >

<!--  This element will be used to include an audio sequence into technical
information.  The model is incomplete pending the Hytime completion.  -->

<!ELEMENT audio-alts  - -  ( audio+ ) >
<!ATTLIST audio-alts
          %a.node-alts;               >

<!--  This element provides the ability to have context sensitive filtering of
audio sequences.  -->

<!ELEMENT video     - -      ( precond*, (%link;)* ) >
<!ATTLIST video
          %a.node;
          external-ptr   IDREF   #REQUIRED  >

<!--  This element will be used to include an video sequence into technical
information.  The model is incomplete pending the Hytime completion.  -->

<!ELEMENT video-alts  - -  ( video+ ) >
<!ATTLIST video-alts
%a.node-alts;               >

<!--  This element provides the ability to have context sensitive filtering of
video sequences.  -->

<!ELEMENT process   - -      ( precond*, (%link;)*, parameter* ) >
<!ATTLIST process
          %a.node;
          external-ptr   CDATA   #REQUIRED    >

<!--  This element is used to reference an external software process.  The
external pointer attribute will point to a location element defined within
Hytime.  The parameter element will provide some expression for passing
parameters to the software process.  -->

<!ELEMENT process-alts  - -  ( process+ ) >
<!ATTLIST process-alts
          %a.node-alts;                   >

<!--  This element provides the ability to have context sensitive filtering of
processes and reduces data redundancy through the referencing capability of
#CONREF.  -->

<!ELEMENT parameter     - -    ( expression ) >
<!ATTLIST parameter
          mode      ( in | out | in-out )    'in' >
```


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```
<!--  This element includes an expression which will be used to create the
parameters required by an external software process. For example: the 1553 bus
on the aircraft might require parameters concerning a given channel to look
up.  The parameter element will contain the channel required by the process.
-->

                        <!-- *****  DIALOGS  ***** -->

<!-- "Dialogs" provide interactivity between the user and the electronic
technical information.  It is sometimes necessary to receive data from the
user in order to present the proper information at the proper time.  "Dialogs"
provide the capability of prompting the user to input a response ("fillin"),
select a choice from a set of alternatives ("menu"), or to select items from
within a text, table or graphic ("selection").  -->

<!ELEMENT dialog      - -   ( precond*, (%link;)*, ( %text;)?, ( %dialog; |
                              fillin | menu |  selection )+ ) >
<!ATTLIST dialog
          %a.node;  agent      CDATA       #IMPLIED >

<!--  This element defines a "dialog" which provides the capability for user
interaction.  A "dialog" could contain a subdialog, a "fillin", a "menu", a
"selection", or any combination of the four.  It may also contain an optional
text string which would be the title of the composite dialog.  The 'agent'
attribute defines to whom the question is asked ( i.e. a technician or a 1553
Bus ).  -->

<!ELEMENT dialog-alts     - -      ( dialog )+ >
<!ATTLIST dialog-alts
          %a.node-alts; >

<!--  This element provides the ability to have context sensitive filtering of
dialogs and reduces data redundancy through the referencing capability of
#CONREF.  -->

<!ELEMENT fillin  - -   ( (%link;)*, prompt, property, ( %text )?,
                          generic-range?)     >
<!ATTLIST fillin
          id             ID             #IMPLIED
          ref            IDREF          #CONREF  >

<!--  This element defines how a fill-in shall be constructed.  It would
contain a "prompt", a "property", and an optional default value.  The "prompt"
contains the information to be presented to the user.  The property element
identifies the variable which will receive a value from the user's response.
The property element also identifies the value type of the user's response.
The fill-in will be presented to the user according to the value type.  The
optional text element provides the capability for defining a default value for
the fill-in.  The generic range element may be used to provide a range for the
value(s) of the fill-in. -->

<!ELEMENT generic-range  - -  ( set | sequence | num-range ) >

<!ELEMENT num-range  - -  ( low-bound, high-bound ) >

<!ELEMENT low-bound  - -  ( integer | real ) >

<!ELEMENT high-bound  - -  ( integer | real ) >
```


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```
<!--  These elements define two types of range constraints.  If the generic
range contains a set or sequence, then the contents of that set or sequence
become the constraints for the fillin.  If the generic range contains a number
range, then the low and high bounds define the constraints for the fillin. -->

<!ELEMENT menu      - -    ( (%link;)*, prompt, choice+ ) >
<!ATTLIST menu
          id             ID             #IMPLIED
          ref            IDREF          #CONREF
          select         ( single | multiple )  'single' >

<!--  This element defines how a "menu" is built for technical information.
It consists of a "prompt" followed by one or more "choices".  The "select"
attribute allows the author to designate the number of choices that may be
selected by the user.   -->

<!ELEMENT prompt    - -   ( %text; | %graphic; ) >
<!ATTLIST prompt
          id             ID             #IMPLIED
          ref            IDREF          #CONREF >

<!--  This element defines the "prompt" to be displayed to the user for the
presentation of a "fillin" or a "menu".  It allows the prompt to be either a
text string (probably in the form of a question) or a graphic (a picture which
requires an answer).-->

<!ELEMENT choice    - -   ( (%link;)*, ( %text; | %graphic; ), ( assertion+ |
                              %dialog;) ) >
<!ATTLIST choice
          id               ID           #IMPLIED
          ref              IDREF        #CONREF
          default        ( Yes | No )    'No'  >

<!--  This element defines the choices for a menu.  A "choice" contains an
optional link, "text" or "graphic" element followed by an assertion or
"dialog" element.  The "link" can be used for example to reference additional
information on the "choice".  The "text" or "graphic" element will be
displayed to the user as a part of the menu.  The assertion or dialog
identifies the action to be taken if the user selects that choice.  The
default attribute provides a method of indicating whether a choice is
designated as a default for the menu.   -->

<!ELEMENT selection  - -  ( ( (%link;)*, ( assertion+ | %dialog; ) )+,
                            ( text | table | graphic ) ) >
<!ATTLIST selection
          id             ID             #IMPLIED
          ref            IDREF          #CONREF >

<!--  This element provides the capability of creating a special "dialog" that
allows selection within a given picture, text string or table.  The semantics
require a 'link' for each selectable item within the text, table, or graphic
selection.  Each 'link' must have at least one linkend specifying the
selectable element in the text, table, or graphic.  Each link will be paired
with an assertion or dialog specifying the action to be taken if that item is
selected.   -->

<!-- ******************************************************
```


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CONTEXT FILTERING ELEMENT DECLARATIONS

```
     ********************************************************
```

Context dependent filtering provides the capability to present the user with only the information that applies to his specific situation. The precondition and postcondition elements provide the mechanism for context dependent filtering. The precondition element enables the selection of the appropriate information for presentation. The postcondition element enables the recording of presentation events for later filtering.

This mechanism assumes that a state table is maintained at presentation time to represent the current situation. The current situation or state is defined by a set of property value pairs. A property value pair associates a value to a property name. It provides the capability to obtain a value by looking up a property name in the state table. -->

```
<!ELEMENT precond    - -      ( expression ) >
<!ATTLIST precond
           id               ID                    #IMPLIED
           ref              IDREF                 #CONREF  >

<!--  A precondition contains an expression to be evaluated at presentation
time.  The precondition is satisfied if the evaluation results in "true".  -->

<!ELEMENT postcond    - -      ( assertion ) >
<!ATTLIST postcond
            id               ID         #IMPLIED
            ref              IDREF      #CONREF >

<!--  The postcondition contains an assertion which is evaluated whenever the
node containing the postcondition is traversed.  After a NODE has been
presented, the assertion will be evaluated and the appropriate property value
pairs will be asserted.  The most recent assignment will overwrite any
previous value.   -->

<!ENTITY % binop  " eq | ne | lt | gt | le | ge | and | or | xor | concat |
                    substring | append | plus | minus | times | divide |
                  idivide | exponent | mod | remove | union | intersect |
                    set-diff | member | subset | disjoint | add |
                      subsequence " >

<!--  The binary operation entity enumerates all of the possible binary
operators which may be used within an expression.  -->

<!ENTITY % unop  " not | empty | size | head | tail | neg | remove | trunc |
                   float | index | undef| max | min" >

<!--  The unary operation entity enumerates all of the unary operators which
may be used within an expression.  -->

<!ENTITY  % value  " boolean | string | sequence | set | real | integer |
                     nil " >

<!--  This entity enumerates the legal value types which properties may
contain.    -->

<!ELEMENT expression   - - ( ( expression, (%binop;),expression) | (( %unop;),
```


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```
                                expression ) | property | (%value;) ) >
<!ATTLIST expression
                      id        ID        #IMPLIED
                              ref       IDREF      #CONREF  >

<!--  The expression element contains one of four types of subexpressions: a
binary operation between two expressions, a unary operation upon an
expression, a property, or a value. If the expression contains a binary or
unary operation, the value is defined by the semantic rules specified later in
this section.  If the expression contains a property, the value of the
expression is obtained by looking up the current value of the property in the
state table. If the expression contains a value, that value is returned as the
result.  -->

<!ELEMENT assertion  - -  ( property, expression ) >
<!ATTLIST assertion
              id            ID             #IMPLIED
              ref           IDREF          #CONREF  >

<!--  The assertion element provides the mechanism for pairing a value with a
property.  The semantics of assertions vary from postconditions.  When an
assertion is present in a node, the presentation rules for that content
specific element will determine whether the assertion is evaluated. -->

<!ELEMENT ( eq, ne, lt, gt, le, ge, and, or, xor, concat,
            substring, plus, minus, times, divide, idivide,
            exponent, mod, union, intersect, set-diff,
            member, subset,  disjoint, append, subsequence,
            not, empty, size, head, tail, neg, trunc, float,
            undef, max, min )                                    - o   EMPTY >

<!ELEMENT add           - o          ( index-value )? >

<!ELEMENT remove        - o          ( index-value, index-value? )? >

<!ELEMENT index         - o          ( index-value, index-value? ) >

<!ELEMENT index-value    - o          ( #PCDATA )  >

<!--  The above elements are used to identify the operators which may be
applied in an expression.    -->

<!ELEMENT property       - -           ( #PCDATA ) >
<!ATTLIST property
          id                  ID        #IMPLIED
          ref                 IDREF     #CONREF
          type                CDATA     #REQUIRED
          value-type          CDATA     'general'
          dialog-ref          IDREF     #IMPLIED>

<!--  The property element contains parsable character data which represents
the property (variable) name.  The value of a property may be obtained by
finding the current value associated with the property name in the state
table.

The 'type' attribute contains a character string which may be used by the
author to identify different property classes.  The 'value-type' attribute is
used to denote the allowable data types which may be assigned to the property.
```


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```
The current legal values for 'value-type' are any combination of the
following: "boolean", "integer", "real", "set", "sequence", "string", and
"general".  The 'dialog-ref' attribute will hold the IDREF of a "dialog" or
"process" element which will acquire a value for the property, if "property"
is undefined (i.e., equal to "nil") at presentation time.  -->

<!ELEMENT ( boolean, string, real, integer ) - -   ( #PCDATA ) >

<!--  These elements define the values boolean, string, real, and integer to
be character data.  -->

<!ELEMENT ( set, sequence )      - -    ( %value; )* >

<!--  These element are used to define a set or sequence as being zero, one or
more values.  -->

<!ELEMENT nil             - o      EMPTY >

<!-- This element signifies that the associated property has a value
of undefined, while the content is EMPTY.  Property's of any type can take on
the “nil” value.  -->

<!-- ************************************************************

                  Semantics of expression operations

     ************************************************************
```

Listed below are the value-types allowed in the generic layer and the valid operators under each value-type, and the semantics of each operation including the return value-type. Unspecified cases shall automatically be considered errors.

OPERATIONS WHICH APPLY TO MULTIPLE DATA TYPES:

```
Operation:     <eq|ne>
Form:          <value><eq|ne><value>
Return Value:  <boolean>
Meaning:       If both operands are the same value-type (or both are numbers )
               then the return value is dependent upon what eq|ne means for
               that value-type. If the operands are of different types, the
               return value is 'False'.

Operation:     <size>
Form:          <size><string>
               <size><sequence>
               <size><set>
Return Value:  <boolean>
Meaning:       An integer value which is the length of the <string> or the
               number of values in the <set|sequence>. For <set|sequence> this
               number represents the members of the <set|sequence>. It does
               not count the elements which are members of an included
               <set|sequence>.

Operation:     <empty>
Form:          <empty><string>
               <empty><set>
```


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	<empty><sequence>
Return Value:	<boolean>Meaning: True if the string, set, or sequence is empty. False otherwise. Logically equivalent to size(<..>) = 0.

Operation:	<index>
Form:	<index><string> <index><sequence>
Return Value:	<string> | <sequence>
Meaning:	The index operator can have one or two index-values in its SGML content. An index-value is a signed integer value. Its meaning is dependent upon its sign. A positive value means an index position counted from the beginning of the <string|sequence>. A negative number means an index position counted back from the end of the <string|sequence>. A zero means the end of the string.

Operation:	<add>
Form:	<set><add><value> <sequence><add><value>
Return Value:	<set> | <sequence>
Meaning:	For a set, add simply means make a new set which has all the members of the old set plus <value>. For a sequence the add operator shall have an index-value as described above for the index operation. The <value> will be inserted before the position pointed to by the index position. If no index-value is given the <value> is added at the end of the sequence.

Operation:	<remove>
Form:	<set><remove><value> <sequence><remove><value> <remove><sequence> <remove><string>
Return Value:	<set> | <sequence> | <string>
Meaning:	For a <set> remove returns a <set> with <value> removed. For a <sequence>using the binary operand form it returns a <sequence> which has the first instance of <value> removed. For a <sequence> or <string> as a unary operator remove must contain an index-value which refers to the position from which the character in the <string> is to be removed or the value in the <sequence> is to be removed. The new string or sequence will be the old one up to but not including the index position concatenating with the old one after the index position.

Operation:	<member>
Form:	<value><member><set> <value><member><sequence>
Return Value:	<boolean>
Meaning:	True if <set|sequence> contains an member who is equal to <value>. False otherwise. This is not a recursive search on any <set|sequence> that might be part of the <set|sequence>.

BOOLEAN OPERATIONS:

Operation:	<or>
Form:	<boolean><or><boolean>
Return Value:	<boolean>
Meaning:	The boolean or function.

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```
Operation:      <and>
Form:           <boolean><and><boolean>
Return Value:   <boolean>
Meaning:        The boolean and function.

Operation:      <xor>
Form:           <boolean><xor><boolean>
Return Value:   <boolean>
Meaning:        The boolean xor function.

Operation:      <not>
Form:           <not><boolean>
Return Value:   <boolean>
Meaning:        The boolean not function.
```

STRING OPERATIONS:

```
Operation:     <concat>
Form:          <string><concat><string>
Return Value:  <string>
Meaning:       The return value is a new string which is equal to the first
               string with the second string concatenated to the end of it.

Operation:     <empty>
Form:          <empty><string>
Return Value:  <boolean>
Meaning:       True if the string is empty ( size = 0 ). False otherwise.
               This is equivalent to size ( <string> = 0 ).

Operation:     <substring>
Form:          <string><substring><string>
Return Value:  <boolean>
Meaning:       True if the first string is a substring of the second string.
               False otherwise.
```

SEQUENCE OPERATIONS

```
Operation:     <append>
Form:          <sequence><append><sequence>
Return Value:  <sequence>
Meaning:       A new sequence equal to the first sequence with the second
               sequence appended to the end.

Operation:     <subsequence>
Form:          <sequence><subsequence><sequence>
Return Value:  <boolean>
Meaning:       True if the first sequence is a subsequence of the second.
               False otherwise.

Operation:     <head>
Form:          <head><sequence>
Return Value:  <value>
Meaning:       Returns the first element in sequence.

Operation:     <tail>
Form:          <tail><sequence>
Return Value:  <sequence>
Meaning:       Returns a sequence with the first element removed.
```


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SET OPERATIONS

```
Operation:     <union>
Form:          <set><union><set>
Return Value:  <set>
Meaning:       A new set which is the union of the two sets.

Operation:     <intersect>
Form:          <set><intersect><set>
Return Value:  <set>
Meaning:       A new set which is the intersection of the two sets.

Operation:     <set-diff>
Form:          <set><set-diff><set>
Return Value:  <set>
Meaning:       A new set which is the difference of the two sets.

Operation:     <disjoint>
Form:          <set><disjoint><set>
Return Value:  <boolean>
Meaning:       True if the intersection of the two sets is empty.  False
               otherwise. This is equivalent to empty( <set1> intersect
               <set2>).

Operation:     <subset>
Form:          <set><subset><set>
Return Value:  <boolean>
Meaning:       True if the first set is a subset of the second.  False
               otherwise.
```

NUMBER OPERATIONS

```
Operation:     <gt>
Form:          <integer><gt><integer>
               <integer><gt><real>
               <real><gt><integer>
               <real><gt><real>
Return Value:  <boolean>
Meaning:       True if the first number is greater than the second.  False
               otherwise.

Operation:     <ge>
Form:          <integer><ge><integer>
               <integer><ge><real>
               <real><ge><integer>
               <real><ge><real>
Return Value:  <boolean>
Meaning:       True if the first number is greater than or equal to the
               second. False otherwise.

Operation:     <lt>
Form:          <integer><lt><integer>
               <integer><lt><real>
               <real><lt><integer>
               <real><lt><real>
Return Value:  <boolean>

Meaning:       True if the first number is less than the second. False
```


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```
                otherwise.

Operation:      <le>
Form:           <integer><le><integer>
                <integer><le><real>
                <real><le><integer>
                <real><le><real>
Return Value:   <boolean>
Meaning:        True if the first number is less than or equal to the second.
                False otherwise.

Operation:      <plus>
Form:           <integer><plus><integer>
                <integer><plus><real>
                <real><plus><integer>
                <real><plus><real>
Return Value:   <integer> | <real>
Meaning:        Return the value of the first number plus the second number.
                The return value is a real unless both numbers are integers.

Operation:      <minus>
Form:           <integer><minus><integer>
                <integer><minus><real>
                <real><minus><integer>
                <real><minus><real>
Return Value:   <integer> | <real>
Meaning:        Return the value of the first number minus the second number.
                The return value is a real unless both numbers are integers.

Operation:      <times>
Form:           <integer><times><integer>
                <integer><times><real>
                <real><times><integer>
                <real><times><real>
Return Value:   <integer> | <real>
Meaning:        Return the value of the first number times the second number.
                The return value is a real unless both numbers are integers.

Operation:      <divide>
Form:           <integer><divide><integer>
                <integer><divide><real>
                <real><divide><integer>
                <real><divide><real>
Return Value:   <real>
Meaning:        Return the value of the first number divided by the second
                number. The return value is a real.

Operation:      <idivide>
Form:           <integer><idivide><integer>
                <integer><idivide><real>
                <real><idivide><integer>
                <real><idivide><real>
Return Value:   <integer>
Meaning:        Return the value of the first number divided by the second
                number. The return value is truncated to an integer.

Operation:      <exponent>
Form:           <integer><exponent><integer>
```


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```
                <integer><exponent><real>
                <real><exponent><integer>
                <real><exponent><real>
Return Value:   <integer> | <real>
Meaning:        Return the value of the first number raised to the power of the
                second number. The value is a real unless the first number is
                an integer and the second number is a positive integer.

Operation:      <mod>
Form:           <integer><mod><integer>
Return Value:   <integer>
Meaning:        The return value is equal to the integer remainder after the
                first number is integer-divided by the second. (This is the
                standard definition of the mod operator).

Operation:      <neg>
Form:           <neg><integer>
                <neg><real>
Return Value:   <integer> | <real>
Meaning:        The return value is the negative of the number. It is an
                integer if the number is an integer, and real if the number is
                a real.

Operation:      <trunc>
Form:           <trunc><integer>
                <trunc><real>
Return Value:   <integer>
Meaning:        The return value is the number truncated to be an integer.

Operation:      <float>
Form:           <float><integer>
                <float><real>
Return Value:   <real>
Meaning:        The return value is the number converted to a real number
                value.

-->
```


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A CONTENT SPECIFIC DOCUMENT TYPE DEFINITION (DTD)

B.1 SCOPE.

B.1.1 Scope. The DTD within this appendix provides the structure and content of documents prepared in accordance with this specification. Unless otherwise specified by the procuring activity, this Appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

B.2 APPLICABLE DOCUMENTS.

B.2.1 Government documents.

B.2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

ISO 8879	Information Processing - Text and Office Systems -Standard Generalized Markup Language (SGML)
ISO 10744	Information Technology - Hypermedia/Time-based Document Structuring Language (Hytime)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

B.3 A CONTENT SPECIFIC DOCUMENT TYPE DEFINITION.

B.3.1 Use of SGML. The markup tags described herein are based on the rules outlined in ISO 8879. All data to be delivered digitally in accordance with this specification shall be tagged using the DTD in this section and the generic DTD found in Appendix A.

B.3.2 Template document type. The DTD for this specification is as follows:

```
<!DOCTYPE  techinfo  [

<!--  ************************************************

          IETM CONTENT DATA MODEL       Version 6.1
          Content Specific DTD       1 October 1992

      ************************************************  -->

<!--  This document contains a content specific DTD for O-level maintenance.
The creation of a content specific DTD represents the second layer of the CDM.
It identifies all the content specific elements and their relationships for a
```


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given application. In this instance, the application happens to be the display of organizational level data to the technician.

The DTD employs the characteristics defined by the templates of the "Generic Layer." The use of the generic layer primitives means that we do not have to redefine the text, table, graphic, audio, video, or process elements within this document.

This document breaks down O-level data into a hierarchy based upon the system/subsystem structure of the weapon system. It identifies four different types of information which may be referenced within the document. They are; procedural, descriptive, parts, and fault information. Each type of information is referenced by the system where it is most appropriate.

```
     ********************************************************

                    PUBLIC ENTITY DECLARATIONS

     ********************************************************

<!ENTITY % dietmdb-b PUBLIC "-//USA-DOD//DTD Content Data Model Content
Specific Layer//EN">  -->

<!ENTITY  % dietmdb-a PUBLIC "-//USA-DOD//DTD Content Data Model Generic
Layer//EN">
%dietmdb-a;

<!--  This entity includes the public identifier for the generic layer of the
CDM.  It provides access to the template, primitive, user-interaction, and
filtering elements within the generic layer.  -->

<!-- The following entities are used to refer to the elements used in this
content specific DTD. They use the node and node alt templates from the CDM
generic layer. These entities are here because of the top down methodology of
SGML.  By defining the entities at the beginning of the DTD, any element below
this point can use the entity declarations.-->

<!ENTITY % sub-prims  " %text; | %table; | %graphic; | %audio; | %video; |
%process; | %dialog; " >
<!ENTITY  % system      "system | system-alts" >
<!ENTITY  % descinfo    "descinfo | descinfo-alts" >
<!ENTITY  % task        "task | task-alts" >
<!ENTITY  % reqcond     "reqcond | reqcond-alts"  >
<!ENTITY  % input       "input | input-alts"  >
<!ENTITY  % person      "person | person-alts"  >
<!ENTITY  % equip       "equip | equip-alts"  >
<!ENTITY  % refmat      "refmat | refmat-alts"  >
<!ENTITY  % expend      "expend | expend-alts"  >
<!ENTITY  % consum      "consum | consum-alts"  >
<!ENTITY  % alert       "alert | alert-alts" >
<!ENTITY  % step        "step | step-alts" >
<!ENTITY  % follow-on   "follow-on | follow-on-alts"  >
<!ENTITY  % partinfo    "partinfo | partinfo-alts"  >
<!ENTITY  % partbase    "partbase | partbase-alts"  >
<!ENTITY  % connection  "connection | connection-alts" >
<!ENTITY  % attach-part "attach-part | attach-part-alts" >
<!ENTITY  % location    "location | location-alts" >
<!ENTITY  % faultinf    "faultinf | faultinf-alts " >
<!ENTITY  % test        "test | test-alts" >
```


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```
<!ENTITY  % outcome     "outcome | outcome-alts" >
<!ENTITY  % fltstate    "fltstate | fltstate-alts" >
<!ENTITY  % fault       "fault | fault-alts" >
<!ENTITY  % rect        "rect | rect-alts" >
<!ENTITY  % para        "para | para-alts" >

<!--  ************************************************

                    Techinfo Declaration

      ************************************************  -->


<!ELEMENT techinfo   - -   ( version+, (%system;)+ )  >
<!ATTLIST techinfo
          %a.node;
          %a.root; >

<!--  This element declaration represents the top layer of the information
contained in the DTD.  The content model contains the top level system, such as
"F-15", "M-1" or "F/A-18".  The hierarchy begins at this level.  -->

<!--  ************************************************

                      System Declaration

      ************************************************  -->

<!--  The system element defines the vehicle/system/subsystem/ subassembly
hierarchy for the weapon system.  A system element must be created for any
component (ie., vehicle, system, subsystem, subassembly) which has associated
technical information (ie., descriptive, procedural, fault, or part
information). -->

<!ELEMENT system  - -  ( precond*, (%link;)*, (%system;)*, (%descinfo;)*,
                       (%task;)*, (%partinfo;)*, (%faultinf;)*  )  >
<!ATTLIST system
          %a.node;
          version       IDREF       #REQUIRED
          status        ( u | a )         'a'  >

<!--  The system element employs the 'NODE' template from the generic layer.  A
"system" contains a list of preconditions which define the elements
applicability, relational links to other elements, sub-system elements, and
descriptive, task, part, and fault information about the system.  -->

<!ELEMENT system-alts  - -  ( system )+  >
<!ATTLIST system-alts
          %a.node-alts; >

<!--  This element employs the 'NODE ALTS' template from the generic layer to
facilitate the context filtering at the system level.  -->


<!--  ************************************************

                    Version Declaration

      ************************************************  -->
```


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```
<!--  The following declaration is an attempt at controlling different versions
of technical information in the database.  A more robust description of how the
version element will work is to follow.  -->

<!ELEMENT  version    - -      ( %text; )?  >
<!ATTLIST  version
          %a.node;
          revision      NMTOKEN      #REQUIRED
          revdate       NUMBER       #REQUIRED
          changeno      NUMBER       #REQUIRED
          chgdate       NUMBERS      #REQUIRED
          deleted       NMTOKENS     #IMPLIED   >

<!--  ************************************************

               Descriptive Information  Declaration

      ************************************************  -->

<!--  The element "descinfo" is used to define general purpose, non-procedural,
narrative information such as theory of operation, schematics, etc which are
associated with a system component.  "Descinfo" is  very flexible.  It can be
used to describe any arbitrary, hierarchical hypertext-like node containing
sub-paragraphs ("para-seq"), ("text", "table", "graphic", "audio", "video",
"process"), user interaction instructions ("dialog"), and postcondition
properties ("postcond") which are asserted whenever the "descinfo" is read.
-->

<!ELEMENT descinfo     - -  ( precond*, (%link;)*, para-seq,  postcond*  ) >
<!ATTLIST descinfo
          %a.node;
          version     IDREF       #REQUIRED
          status      ( u | a )   'a' >

<!--  The descinfo element employs the 'NODE' template from the generic layer.
"Descinfo" contains a list of preconditions which define the element's
applicability, relational links to other elements, paragraph sequences, and a
list of postconditions which may change the state of the system.   -->

<!ELEMENT descinfo-alts  - -  ( descinfo )+  >
<!ATTLIST descinfo-alts
          %a.node-alts; >

<!--  This element employs the 'NODE ALTS' template from the generic layer to
facilitate the context filtering of descriptive information.  -->


<!--  ************************************************

                       Para Declaration

      ************************************************  -->

<!ELEMENT  para   - -  ( precond*, (%link;)*, (%sub-prims;)+, para-seq?,
                         postcond* ) >
<!ATTLIST  para
          %a.node;
          version       IDREF       #REQUIRED
          status        ( u | a )         'a' >
```


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```
<!--  The "para" element employs the NODE template.  It defines the information
which may be contained within the descriptive information as any primitive
element defined in the generic layer.  -->

<!ELEMENT  para-alts - -  ( para )+  >
<!ATTLIST  para-alts
           %a.node-alts;    >

<!--  This element employs the 'NODE ALTS' template from the generic layer to
facilitate the context filtering of paragraph information.  -->

<!ELEMENT  para-seq   - -  ( %descinfo; | para | para-alts |if-para |
                           loop-para )+ >
<!ATTLIST  para-seq
           %a.node-seq;   >

<!--  This element employs the 'NODE SEQ' template from the generic layer.  It
provides the capability to create sequences of paras.    -->

<!ELEMENT if-para      - -  ( expression, para-seq, para-seq? ) >
<!ATTLIST if-para
          %a.if-node;   >

<!--  This element employs the 'IF NODE' template from the generic layer.  It
allows conditional selection of paras depending on a precondition.  -->

<!ELEMENT loop-para   - -   ( assertion?, expression, assertion?, para-seq ) >
<!ATTLIST loop-para
          %a.loop-node;  >

<!--  This element employs the 'LOOP NODE' template from the generic layer.  It
provides the capability of looping through a sequence of paras.  -->

<!-- ***********************************************

                     Task Declaration

      ***********************************************  -->

<!--  The element "task" defines a maintenance procedure, such as a removal,
repair, replacement, test, adjustment, etc. associated with a "system"
component.  -->


<!ELEMENT task     - -  ( precond*, (%link;)*, (%input;)*,
                          step-seq, (%follow-on;)*, postcond* ) >
<!ATTLIST task
          %a.node;
          version        IDREF                    #REQUIRED
          status         ( u | a )                  'a'
          esttime        NUTOKEN                  #IMPLIED
          operability    CDATA                    #IMPLIED
          servicedes     CDATA                    #IMPLIED  >

<!--  The "task" element employs the 'NODE' template from the generic layer.  A
"task" element contains a list of preconditions which define the task's
applicability, relational links to other information elements and input
conditions for beginning the task, precautionary messages (i.e., warnings,
cautions and notes), a sequence of procedural steps, a list of follow-on
```


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```
conditions which must be accomplished sometime following the completion of the
task, and a list of postconditions which define any state changes to be made
after the task is accomplished.    -->

<!ELEMENT task-alts  - -  ( task )+  >
<!ATTLIST task-alts
          %a.node-alts; >

<!--  This element employs the 'NODE ALTS' template from the generic layer to
facilitate the context filtering of tasks.  -->

<!--  ************************************************

                        Input Declaration

      ************************************************  -->

<!--  The input element identifies all the set-up conditions which must be met
prior to beginning a task.  -->

<!ELEMENT input   - -  ( precond*, (%link;)*, (%alert;)*, (%reqcond;)*,
                       (%person;)+, (%refmat;)*, (%equip;)*, (%expend;)*,
                       (%consum;)* ) >
<!ATTLIST input
          %a.node;
          version     IDREF       #REQUIRED
          status      ( u | a )    'a' >

<!--  The "input" element employs the 'NODE' template from the generic layer.
An "input" contains applicability preconditions, relational links to other
elements, and the personnel, consumables, equipment and required conditions for
accomplishing the task.  -->

<!ELEMENT input-alts       - -     ( input )+ >
<!ATTLIST input-alts
          %a.node-alts; >

<!--  This element employs the 'NODE ALTS' template from the generic layer to
facilitate the context filtering of input conditions.  -->

<!--  ************************************************

                   Required Condition Declaration

      ************************************************  -->

<!--  A required condition (RECOND) identifies a maintenance condition (eg.,
aircraft safe for maintenance ), which must be satisfied before beginning a
task.  It also identifies the task(s) or step(s) which accomplish the required
condition if it is not satisfied. -->

<!ELEMENT reqcond     - -  ( precond*, (%link;)*, (%text;)?,  ( expression,
                           ( %task; | %step; ), assertion* ), postcond* ) >
<!ATTLIST reqcond
          %a.node;
          version        IDREF      #REQUIRED
          status         ( u | a)    'a' >
```


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```
<!--  The "reqcond" element employs the 'NODE' template from the generic layer.
A "reqcond" contains a set of preconditions which define the required
maintenance condition's applicability, relational links, an optional text
element which describes the maintenance condition, a list of task(s) or step(s)
which provide instructions for accomplishing the maintenance condition, and a
set of postconditions which define the state changes to be made once the
maintenance condition is accomplished.   -->

<!ELEMENT reqcond-alts - -     ( reqcond )+ >

<!ATTLIST reqcond-alts
         %a.node-alts;   >

<!--  This element employs the 'NODE ALTS' template from the generic layer to
facilitate the context filtering of required conditions.  -->

<!--  ************************************************

                  Reformat and Expend Declarations

      ************************************************  -->

<!--  The following elements identify reference material and expendables for a
task.    -->

<!ELEMENT  refmat  - -  ( precond*, (%link;)*, (%text;)? ) >
<!ATTLIST  refmat
        %a.node;
        version      IDREF          #REQUIRED
        status       ( u | a )            'a'
        desig        CDATA          #REQUIRED >

<!ELEMENT  refmat-alts   - -     ( refmat )+ >
<!ATTLIST  refmat-alts
                       %a.node-alts;   >

<!ELEMENT  expend  - -  ( precond*, (%link;)*, (%partbase;)?, (%consum;)* ) >
<!ATTLIST  expend
                       %a.node;
                       version            IDREF          #REQUIRED
                       status       ( u | a )            'a'
                       quantity     CDATA        #REQUIRED >

<!ELEMENT  expend-alts    - -    ( expend )+  >
<!ATTLIST  expend-alts
                       %a.node-alts;    >

<!--  ************************************************

                       Person Declaration

      ************************************************  -->

<!--  This element is used to identify the personnel requirements for a given
task.  The 'type' attribute will be used to identify the kind of technician
required.  The 'quantity' attribute identifies the number of that type of
technician required for the task.  -->
```


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```
<!ELEMENT person     - -    ( precond*, (%link;)*, (%text;)? )  >
<!ATTLIST person
          %a.node;
          version       IDREF      #REQUIRED
          status        ( u | a )        'a'
          quantity      CDATA     #IMPLIED  >

<!--  The person element employs the 'NODE' template from the
generic layer.   -->

<!ELEMENT person-alts       - -     ( person )+ >
<!ATTLIST person-alts
          %a.node-alts; >

<!--  This element employs the 'NODE ALTS' template from the generic layer to
facilitate the context filtering of person elements.  -->

<!--  ************************************************

                      Equipment Declaration

      ************************************************  -->

<!--  This element identifies all the support equipment required for the
completion of the task.  -->

<!ELEMENT equip   - -  ( precond*, (%link;)*, (%equip;)*, (%text;)?,
                       (%partbase;)? ) >
<!ATTLIST equip
          %a.node;
          version      IDREF      #REQUIRED
          status       ( u | a )        'a'
          quantity     CDATA     #IMPLIED  >

<!--  The equip element employs the 'NODE' template from the generic layer.  An
"equip" contains applicability preconditions, relational links to other
elements, and any alternate equipment. The quantity attribute identifies the
number of equipment items required to complete the task.  -->

<!ELEMENT equip-alts       - -     ( equip )+ >
<!ATTLIST equip-alts
          %a.node-alts;   >

<!--  This element employs the 'NODE ALTS' template from the generic layer to
facilitate the context filtering of equipment elements.  -->

<!--  ************************************************

                      Consumable Declaration

      ************************************************  -->

<!--  This element identifies all the consumables required for the completion
of the task.  -->


<!ELEMENT consum   - -  ( precond*, (%link;)*, (%partbase;)?, (%consum;)* )  >
<!ATTLIST consum
          %a.node;
```


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```
          version          IDREF     #REQUIRED
          status           ( u | a ) 'a'
          govstd           CDATA     #IMPLIED
          mfgcode          CDATA     #IMPLIED
          milspec          CDATA     #IMPLIED
          quantity         CDATA     #REQUIRED
          unit-of-measure  NMTOKEN   #IMPLIED  >

<!--  The consum element employs the 'NODE' template from the generic layer.  A
"consum" contains applicability preconditions and relational links to other
elements.  The "consum" element contains many attributes which identify what
the consumable is (govstd, mfgcode, milspec), and the amount required
(quantity, unit-of-measure) for accomplishing the task.  -->

<!ELEMENT consum-alts       - -     ( consum )+ >
<!ATTLIST consum-alts
          %a.node-alts; >

<!--  This element employs the 'NODE ALTS' template from the generic layer to
facilitate the context filtering of equipment elements.  -->


<!--  ************************************************

                        Alert Declaration

      ************************************************  -->

<!--  This element identifies an alert that may accompany a task or step.  The
'type' attribute will identify the kind of alert, either Warning, Caution,
Note.  -->

<!ELEMENT alert   - -  ( precond*, (%link;)*, (%text;)+, (%graphic;)* ) >
<!ATTLIST alert
          %a.node;
          version        IDREF      #REQUIRED
          status        ( u | a )         'a' >

<!--  The alert element employs the 'NODE' template from the generic layer.  An
"alert" contains applicability preconditions, relational links, text elements
which make up the content of the alert message, and optional "graphic" icons to
be displayed.  -->

<!ELEMENT alert-alts       - -     ( alert )+ >
<!ATTLIST alert-alts
          %a.node-alts;   >

<!--  This element employs the 'NODE ALTS' template from the generic layer to
facilitate the context filtering of alerts.  -->


<!--  ************************************************

                        Step Declaration

      ************************************************  -->
```


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```
<!--  Steps are the primary component of a maintenance procedure.  They
describe the actions to be performed in order to successfully complete the
task.   -->

<!ELEMENT step     - -  ( precond*, (%link;)*, (%alert;)*, (%sub-prims;)*,
                          step-seq?, postcond* ) >
<!ATTLIST step
          %a.node;
          version       IDREF             #REQUIRED
          status        ( u | a )         'a'
          esttime       NUTOKEN           #IMPLIED  >

<!--  The step element employs the 'NODE' template from the generic layer.  A
"step" contains a list of preconditions which delimit the step's applicability,
relational links, precautionary alerts, an optional sequence of substeps, and a
list of postconditions which define the state changes to be made after the step
is accomplished. -->

<!ELEMENT step-alts       - -     ( step )+ >
<!ATTLIST step-alts
          %a.node-alts;     >

<!--  This element employs the 'NODE ALTS' template from the generic layer to
facilitate the context filtering of steps.   -->

<!ELEMENT step-seq  - -  ( step | step-alts | if-step | loop-step | task |
                           task-alts )+ >
<!ATTLIST step-seq
          %a.node-seq;     >

<!--  This element employs the 'NODE SEQ' template from the generic layer.  It
provides the capability to create sequences of steps.    -->

<!ELEMENT if-step  - -  ( expression, step-seq, step-seq? ) >
<!ATTLIST if-step
           %a.if-node;   >

<!--  This element employs the 'IF NODE' template from the generic layer.  It
allows conditional selection of steps depending on a precondition.  -->

<!ELEMENT loop-step  - -  ( assertion?, expression, assertion?,  step-seq ) >
<!ATTLIST loop-step
          %a.loop-node;    >

<!--  This element employs the 'LOOP NODE' template from the generic layer.  It
provides the capability of looping through a sequence of steps.  -->


<!--  ***********************************************

                    Follow-on Declaration

      ***********************************************  -->

<!--    A follow-on condition is a maintenance condition which must be
accomplished sometime following the completion of a task to clean up or undo
actions performed during the task.  For example, in order to fix a component a
task might require that an access panel be removed.  The panel would then need
to be replaced as a follow-on action.  This task might be performed sometime
```


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```
after the repair task is completed, but not immediately after the repair task.
Other maintenance tasks might be performed in the same area before the
follow-on task is accomplished. -->

<!ELEMENT follow-on    - -  ( precond*, (%link;)*, (%text;)?, ( expression,
                            ( %task; | %step; ), assertion* ), postcond* ) >
<!ATTLIST follow-on
          %a.node;
          version       IDREF       #REQUIRED
          status        ( u | a )         'a' >

<!--  The "follow-on" element employs the 'NODE' template from the generic
layer.  A "follow-on" element contains a set of preconditions which define the
follow-on maintenance condition which must be satisfied, relational links, an
optional text element which describes the follow-on condition, a list of
task(s)/step(s) which provide instructions for accomplishing the follow-on
condition, and a set of postconditions which define the state changes to be
made once the follow-on condition is accomplished.     -->

<!ELEMENT follow-on-alts       - -     ( follow-on )+ >
<!ATTLIST follow-on-alts
          %a.node-alts;  >

<!--  This element employs the 'NODE ALTS' template from the generic layer to
facilitate the context filtering of follow-on elements.  -->


<!--  ********************************************

                    Parts Information

      **********************************************  -->


<!--  "Partinfo" describes the maintainer's view of the part information. Each
"partinfo" element is related to a "partbase."  However, several "partinfo"
items could be related to the same "partbase."  -->

<!ELEMENT partinfo   - -  ( precond*, (%link;)*, (%partinfo;)*, (%partbase;)+,
                           (%connection;)*, (%attach-part;)*, (%text;)?,
                           (%graphic;)*, (%location;)* ) >
<!ATTLIST partinfo
          %a.node;
          version        IDREF     #REQUIRED
          status         ( u | a ) 'a'
          indexnum       NUTOKENS  #IMPLIED
          lru            NUTOKEN   #IMPLIED
          mtbf           CDATA     #IMPLIED
          refdes         NMTOKEN   #IMPLIED
          replvl         CDATA     #IMPLIED
          unitsper       NUTOKEN   #IMPLIED
          usablon        NUTOKENS  #IMPLIED  >

<!--  The "partinfo" element employs the 'NODE' template.  A "partinfo" element
contains a list of preconditions, relational links, and alternate parts
information ( the "partinfo" in the content model).  "Partinfo" also identifies
the components of the part (partbase), any connecting parts (connection),
attaching parts (attach-part), a formal name for the part (text), a picture of
```


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```
the part (graphic), and the location of the part in reference to the weapon
system (location).   -->

<!ELEMENT partinfo-alts       - -     ( partinfo )+ >
<!ATTLIST partinfo-alts
          %a.node-alts;   >

<!--  This element employs the 'NODE ALTS' template from the generic layer to
facilitate the context filtering of part information elements.  -->


<!--  ************************************************

                        Partbase Declaration

      ************************************************  -->

<!--  "Partbase" describes the supply system's view of the part information.
It describes the item in terms of its part number ('partnum').  -->

<!ELEMENT partbase     - -  ( precond*, (%link;)*, (%partbase;)*, (%text;)?,
                            (%location;)* ) >
<!ATTLIST partbase
          %a.node;
          version         IDREF       #REQUIRED
          status          ( u | a )   'a'
          cage            NUTOKENS    #REQUIRED
          fsc             CDATA       #REQUIRED
          partnum         CDATA       #REQUIRED
          smr             CDATA       #REQUIRED
          nsn             CDATA       #IMPLIED
          pmic            CDATA       #IMPLIED
          cac             NUTOKEN     #IMPLIED
          qpei            NUTOKEN     #IMPLIED
          hci             (Y1|N1)     "N1"
          lox             (Y2|N2)     "N2"
          esds            (Y3|N3)     "N3"
          qec             (Y4|N4)     "N4"
          magnetic        (Y5|N5)     "N5"  >


<!--  The "partbase" element employs the 'NODE' template from the generic
layer.  It allows for the declaration of preconditions for partbase information
and relational linking to other information from the partbase element.  -->

<!ELEMENT partbase-alts       - -     ( partbase )+ >
<!ATTLIST partbase-alts
          %a.node-alts;   >

<!--  This element employs the 'NODE ALTS' template from the generic layer to
facilitate the context filtering of part base elements.  -->

<!--  ************************************************

           Connecting and Attaching Parts Declaration

      ************************************************  -->

<!ELEMENT connection    - -  ( precond*, (%link;)*, (%partinfo;)+ ) >
```


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```
<!ATTLIST connection
          %a.node;
          version     IDREF        #REQUIRED
          status      ( u | a)     'a' >

<!--  The connection element employs the 'NODE' template.  It defines a
connection between two "partinfo" elements.  -->

<!ELEMENT connection-alts       - -     ( connection )+ >
<!ATTLIST connection-alts
  %a.node-alts;   >

<!--  This element employs the 'NODE ALTS' template from the generic layer to
facilitate the context filtering of connection elements.  -->

<!ELEMENT attach-part  - -    ( precond*, (%link;)*, (%partinfo;)+ ) >
<!ATTLIST attach-part
          %a.node;
          version      IDREF     #REQUIRED
          status       ( u | a )       'a' >

<!--  The attaching part element employs the 'NODE' template.  It defines the
attaching parts for a "partinfo" element.  -->

<!ELEMENT attach-part-alts       - -     ( attach-part )+ >
<!ATTLIST attach-part-alts
          %a.node-alts;   >

<!--  This element employs the 'NODE ALTS' template from the generic layer to
facilitate the context filtering of attach-part elements.  -->

<!--  ****************************************************

                      Location Declaration

      ****************************************************  -->

<!--  The location element provides information for physical assessment. It
will contain x, y, z location(s) for a system with respect to the x, Fuselage
Station (FS), y, Buttock Line (BL), and z, Water Line (WL) reference system.
Where appropriate BL may be replaced by Wing Station (WS).    -->


<!ELEMENT location       - -  ( precond*, link* ) >

<!ATTLIST location
          %a.node;
          version        IDREF            #REQUIRED
          status         ( u | a )         'a'
          location-x     NUTOKENS          #IMPLIED
          location-y     NUTOKENS          #IMPLIED
          location-z     NUTOKENS          #IMPLIED >

<!--  The location element employs the 'NODE' template from the generic layer.
It allows for the declaration of preconditions for a physical location and
relational linking to other information from the location element.  -->

<!ELEMENT location-alts    - -     ( location )+ >
<!ATTLIST location-alts
```


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```
          %a.node-alts;       >

<!--  This element employs the 'NODE ALTS' template from the generic layer to
facilitate the context filtering of location elements.  -->

<!--  ************************************************

                  Fault Information Declaration

      ************************************************  -->

<!--  The "faultinf" element identifies all the fault isolation information
associated with a system.  "Faultinf" can be used to support dynamic
troubleshooting models or static troubleshooting trees.  -->

<!ELEMENT faultinf  - -  ( precond*, (%link;)*, (%test;)+, (%fault;)* ) >
<!ATTLIST faultinf
          %a.node;
          version      IDREF     #REQUIRED
          status      ( u | a )  'a' >

<!--  The faultinf element employs the 'NODE' template.  It contains a list of
preconditions, relational links to other elements, and lists of tests and
faults associated with a system.  -->

<!ELEMENT faultinf-alts    - -    ( faultinf )+  >
<!ATTLIST faultinf-alts
          %a.node-alts; >

<!--  This element employs the 'NODE ALTS' template from the generic layer to
facilitate the context filtering of fault information elements.  -->

<!--  ************************************************

                      Test Declaration

      ************************************************  -->

<!--  The "test" element identifies a prescribed task to perform and is the
usual way of entering the troubleshooting process.  The result of a test is an
outcome; a test will have one or more outcomes.  -->

<!ELEMENT test        - -  ( precond*, (%link;)*, (%task;), (%outcome;)+ ) >
<!ATTLIST test
          %a.node;
          version       IDREF       #REQUIRED
          status        ( u | a )   'a'
          agent         CDATA       #IMPLIED
          RANGE         CDATA       #IMPLIED  >

<!--  This element identifies the task needed to complete the test and all the
possible outcomes as a result of the test.  -->

<!--  The test element employs the 'NODE' template.  It contains a list of
preconditions and relational links to other information.  A "test" element
identifies the task which will accomplish the test.  All the possible outcomes
are contained within the test.  -->

<!ELEMENT test-alts    - -    ( test )+  >
```


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```
<!ATTLIST test-alts
          %a.node-alts;  >

<!--  This element employs the 'NODE ALTS' template from the generic layer to
facilitate the context filtering of tests.  -->

<!--  ****************************************************

                      Outcome Declaration

      ****************************************************  -->

<!--  This element identifies a result of a test.  The precondition list is
evaluated against the result of the test, and the appropriate outcome is
selected.  In a dynamic troubleshooting model, the outcome will contain a fault
state that identifies an implicated or exculpated set of faults.  In a static
troubleshooting model, the outcome will contain another test or a fault.  The
outcome will contain the information necessary for both models, but it will be
up to the diagnostic software to choose the correct path to follow for its
logic.  -->

<!ELEMENT outcome - -  ( precond*, (%link;)*, expression,  ( (%fltstate;) |
                       ( ( %test; | %fault; ), (%fltstate;)? ) ) ) >
<!ATTLIST outcome
          %a.node;
          version     IDREF    #REQUIRED
          status     ( u | a )       'a' >

<!--  The outcome element employs the 'NODE' template.  It contains a list of
preconditions, and relational links to other information.  The fault state
element will identify the implicated or exculpated faults for the outcome.  The
test and fault elements identify the next step in a static fault tree.  -->

<!ELEMENT outcome-alts    - -    ( outcome )+  >
<!ATTLIST outcome-alts
          %a.node-alts;   >

<!--  This element employs the 'NODE ALTS' template from the generic layer to
facilitate the context filtering of outcomes. -->

<!--  ****************************************************

                     Fault state Declaration

      ****************************************************  -->

<!--  The "fltstate" element identifies a set of implicated or exculpated
faults.  Implicated faults are faults suspected of being bad; exculpated faults
are faults known to be good.  Each implicated fault will have an associated
weight based on its likelihood of causing the discrepancy.  -->

<!ELEMENT fltstate  - -  ( precond*, (%link;)*, (%fault;)+ ) >
<!ATTLIST fltstate
          %a.node;
          version        IDREF          #REQUIRED
          status         ( u | a )      'a'
          weight         NUTOKENS       #IMPLIED >
```


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```
<!--  The fltstate element employs the 'NODE' template.  It contains a list of
preconditions, and relational links to other appropriate information.  The
'type' attribute will designate whether the list of faults are "implicated" or
"exculpated." -->

<!ELEMENT fltstate-alts    - -    ( fltstate )+  >
<!ATTLIST fltstate-alts
          %a.node-alts; >

<!--  This element employs the 'NODE ALTS' template from the generic layer to
facilitate the context filtering of fault states.  -->



<!--  ****************************************************

                        Fault Declaration

      ****************************************************  -->

<!--  The "fault" element identifies the cause of a discrepancy on the weapon
system.  The fault will identify the appropriate rectification to correct the
discrepancy.  When transitioning between maintenance levels the fltstate
element is used.  -->

<!ELEMENT fault   - -  ( precond*, (%link;)*, ( %rect; | %fltstate; )+,
                       (%system;)+ ) >
<!ATTLIST fault
          %a.node;
          version     IDREF     #REQUIRED
          status     ( u | a )  'a'
          mtbf        CDATA     #IMPLIED  >

<!--  The fault element employs the 'NODE' template.  It contains a list of
preconditions, and relational links to other appropriate information.  The
rectifications contain tasks which will correct the discrepancy.  The system
and part information elements will create a back link to the part that has
failed.  The "fltstate" represents the system at the next level of maintenance.
-->

<!ELEMENT fault-alts    - -    ( fault )+  >
<!ATTLIST fault-alts
          %a.node-alts;  >

<!--  This element employs the 'NODE ALTS' template from the generic layer to
facilitate the context filtering of faults.  -->

<!--  ****************************************************

                    Rectification Declaration

      ****************************************************  -->

<!--  The "rect" element identifies the prescribed task that will repair the
fault causing the discrepancy and all other faults that could be fixed by the
rectification.  Upon completion of the task, a test is performed to verify the
effect of the rectification.   -->

<!ELEMENT rect    - -  ( precond*, (%link;)*, (%task;)+, (%fault;)+,
```


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```
                        (%system;),  (%test;)* ) >
<!ATTLIST rect
          %a.node;
          version      IDREF             #REQUIRED
          status      ( u | a )          'a'
          action      ( swap | maint )   #REQUIRED
          agent        CDATA             #IMPLIED >

<!--  The rect element employs the 'NODE' template.  It contains a list of
preconditions, and relational links to other appropriate information.  The
"system" element provides a reference to the system which will be repaired by
the rectification.  The test element identifies all check-out tests required
before completing the maintenance session.  -->

<!ELEMENT rect-alts    - -    ( rect )+  >
<!ATTLIST rect-alts
          %a.node-alts;    >

<!--  This element employs the 'NODE ALTS' template from the generic layer to
facilitate the context filtering of rectifications.  -->
```


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GENERIC LAYER
TAG SET DESCRIPTIONS

C.1 SCOPE.

C.1 Scope. This appendix provides the detailed description of the elements and attributes to be included in an IETMDB. It is formulated as a description of possible tags or names for components in an IETMDB whose structure is defined by the generic layer DTD specified within Appendix A of this specification. Unless otherwise specified by the procuring activity, this Appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

C.2 APPLICABLE DOCUMENTS.

C.2.1 Government documents.

C.2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

ISO 8879	Information Processing - Text and Office Systems - Standard Generalized Markup Language (SGML)
ISO 10744:1992	Information Technology - Hypermedia/Time-based Document Structuring Language (HyTime)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

C.3 GENERIC LAYER TAG SET DESCRIPTIONS.

C.3.1 Use of SGML. The markup tags described herein conform to rules defined in ISO 8879.

C.3.2 Tag Set Descriptions. Data elements shall be defined in accordance with the tag set descriptions included below (see 3.1.2).

**** Please note, several entries are not complete. Awaiting response ***

ADD **Addition**

Descriptor: Element
Format: N/A
Template Used: N/A

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Description: Adds two values. For a **set,** add simply means make a new set which has all the members of the old set plus **value**. For a **sequence** the **add** operator shall have an **index-value** as described below for the index operation. The **value** will be inserted before the position pointed to by the **index** position. If no **index-value** is given the **value** is added at the end of the **sequence**.

ANCHROLE **Anchor**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Identifies the type of primitive or element that the **link** is pointing to.

AGENT **Dialog Agent**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Used in the **dialog** element, this attribute defines to whom the question is asked. The value of this attribute contains character data which identifies the person or computer to whom the dialog should be presented. The default value is '**human**'.

AGGLOC

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description:

AGGTRAV

Descriptor: Attribute Format: NAMES
Template Used: N/A

Description:

ASSERTION **Assertion**

Descriptor: Element Format: N/A
Template Used: N/A

Description: This element is used to make an assertion from within the content model of an application specific element. Whenever an assertion appears in an element's content model, there shall be set of semantic rules describing when the assertion is to be evaluated. For example, under required conditions the assertion is only evaluated when the user decides to skip a task reference.

AUDIO **Audio Sequence**

Descriptor: Element Format: N/A

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Template Used: Node, Node alts

Description: This element is used to hold an audio sequence.

AUDIO-ALTS **Audio Alternatives**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allows for context-sensitive filtering of audio information.

BINOP **Binary Operation**

Descriptor: Entity Format: N/A
Template Used: N/A

Description: This entity enumerates all of the possible binary operators which may be used within an expression. This element must contain one of the following elements: **eq**, **ne**, **lt**, **gt**, **le**, **ge**, **and**, **or**, **xor**, **concat**, **substring**, **append**, **plus**, **minus**, **times**, **divide**, **idivide**, **exponent**, **mod**, **remove**, **union**, **intersect**, **set-diff**, **member**, **subset**, **disjoint**, **add**, **subsequence**.

BOSLEVEL

Descriptor: Attribute Format:
Template Used: N/A

Description:

CDM **CDM Template Type**

Descriptor: Attribute Format: NAME
Template Used: N/A

Description: Used in all element declarations, to identify the generic template which the element follows. The attribute's value is a fixed default value (ie. cannot be changed by entry of another value). It is set to **'node'** if the element follows the '**node**' template. It is set to '**node-alts**' if the element follows the '**node alts**' template. It is set to '**node-seq**' if the element follows the '**node seq**' template. It is set to '**if-node**' if the element follows the '**if node**' template. It is set to '**loop-node**' if the element follows the '**loop node**' template.

CHOICE **Choice**

Descriptor: Element Format: N/A
Template Used: N/A

Description: This element defines a choice in a menu. Choices consist of a piece of text or a graphic to be displayed. Once the user selects a choice from a menu, the presentation system will either

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assert some postcondition or will branch to another **dialog** (which could contain another **menu**, **fillin** or **selection**).

CODING **Coding**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Used by the **grphprim** element, this attribute identifies the particular storage type of the current graphic file (e.g. IGES, CGM). The default value is '**cgmbin**'.

COLHDDEF **Column Header Definition**

Descriptor: Element Format: N/A
Template Used: N/A

Description: This element defines a column header for a specific column of tabular information.

COLNUM **Column Number**

Descriptor: Attribute Format: NUTOKEN
Template Used: N/A

Description: Used by the **colhddef** and **entry** elements, the value of this attribute consists of the column number of a table.

DEFAULT **Default Indicator**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Used by the **choice** element, this attribute contains an enumerated list with values of either a '**Yes**' or '**No**'. The **default** attribute provides a method of indicating whether a **choice** is designated as a default for the **menu**. The default value for this attribute is '**No**'.

DIALOG **User Interactive Dialogs**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: This element provides the capability for user interaction. A **dialog** could contain a sub**dialog**, **fillin**, **menu**, **selection**, or any combination of the four. It may also contain an optional text string which would be the title of the composite dialog.

DIALOG-ALTS **User Interactive Dialogs Alternatives**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

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Description: Allows for context-sensitive filtering of "DIALOG".

DIALOG-REF **Dialog Reference**

Descriptor: Attribute Format: IDREF
Template Used: N/A

Description: Used in the **property** element, this attribute contains the ID of either a **dialog** element or a **process** element which will acquire a value for the **property**, if **property** is undefined (ie., equal to '**nil**') at presentation time.

DOCORSUB

Descriptor: Attribute Format:
Template Used: N/A

Description:

DTDORLPD

Descriptor: Attribute Format: NAMES
Template Used: N/A

Description:

ENDTERMS

Descriptor: Attribute Format: IDREFS
Template Used: N/A

Description:

ENDTYPES **Endtypes**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Indicates the category of data being linked.

ENTRY **Column Entry Definition**

Descriptor: Element Format: N/A
Template Used: N/A

Description: This element defines an entry for a cell in a table. An **entry** is a piece of text and a column number.

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EXPRESSION **Expression**

Descriptor: Element Format: N/A
Template Used: N/A

Description: The **expression** element provides the capability to create mathematical expressions to be used for **preconditions** and **postconditions**. There can be one of four types of expressions: a binary operation between two expressions, a unary operation with an expression, a property, or a value.

EXTERNAL-PTR **External Process Pointer**

Descriptor: Attribute Format: IDREF
Template Used: N/A

Description: Used in the **audio**, **video**, and **process** elements, this attribute is a pointer which points to an external file. The external file shall contain the appropriate audio, video, or software process that will present to the user a multimedia event.

EXTRA

Descriptor: Attribute Format: NAMES
Template Used: N/A

Description:

FILLIN **Fill In The Blank Question**

Descriptor: Element Format: N/A
Template Used: N/A

Description: This element defines a fill in the blank question. It will contain a **prompt**, a **property**, and an optional default value. The **prompt** contains the question to be presented to the user. The **property** element identifies the variable which will receive a value from the user's response. The **property** element also identifies the legal value type of the user's response. The **fillin** will be presented to the user according to the value type.

GENERIC-RANGE **Generic Range**

Descriptor: Element Format: N/A
Template Used: N/A

Description: This element provides a mechanism for defining valid range checking for **fillin** elements. The element may identify a maximum and minimum for numeric entries or a set of valid values that may be entered for an alpha numeric entry.

GRAPHIC **Graphic**

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Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: This element represents a composite graphic which is made up of graphic primitives (**grphprim**) or other graphic components (**graphic**).

GRAPHIC-ALTS **Graphic Alternatives**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allows for context-sensitive filtering of “GRAPHIC”.

GRPHPRIM **Graphic Primitives**

Descriptor: Element Format: N/A
Template Used: Node, Node-alts

Description: This element defines a graphic primitive to be a single graphic component which, when combined with other primitives, can become a composite graphic. A graphic primitive references a file that contains the detailed graphic information in some standard (e.g., CGM, IGES, FAX, or DXF graphic codes).

GRPHPRIM-ALTS **Graphic Primitives Alternatives**

Descriptor: Element Format: N/A
Template Used: Node, Node-alts

Description: Allows for context-sensitive filtering of “GRPHPRIM”.

HIGH-BOUND **HighBound**

Descriptor: Element Format: N/A
Template Used: N/A

Description: This element identifies the maximum allowable number for a numeric entry of a **fillin**.

HYLINK **HyTime Link**

Descriptor: Element Format: N/A
Template Used: N/A

Description: This element provides the capability for creating relational links with the data. It employs the HyTime "**ilink**" architectural form (template) and may contain "anchors" called (location elements) to identify two or more **linkends**. The **link** element may contain the name of the relation (e.g., **linkterm**).

HYTIME **Hytime**

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Descriptor: Attribute Format: NAME
Template Used: N/A

Description: Used by the hylink element, this attribute is a fixed default value (ie. cannot be changed by entry of another value). It is set to **'ilink'**.

ID **Identifier**

Descriptor: Attribute Format: ID
Template Used: N/A

Description: Used by elements to hold a unique identifier of a specific element.

IF-NODE **If Node Template**

Descriptor: Entity Format: Template
Template Used: N/A.

Description: The **if node** template provides a method of conditional branching within an interactive sequence. This template uses the same logic as the IF-THEN-ELSE statement in a programming language.

INDEX **Index**

Descriptor: Element Format: N/A
Template Used: N/A

Description: A signed integer value. Its meaning is dependent upon its sign. A positive value means an index position from the beginning of a **string** or **sequence**. A negative number means an index position counted back from the end of the **string** or **sequence**. A zero means the end of the string.

INDEX-VALUE **Index**

Descriptor: Element Format: N/A
Template Used: N/A

Description:

ITEMID **Item Identification**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Used in all **node** elements to identify the components of the system being repaired, as they relate to information elements. The item identification attribute specifies the reference designator(s) or other identifiable designator(s) of the system(s), subassemblies, or parts referred to

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by the element. The permissible values of this attribute are dependant upon the content specific application using this primitive.

INTRA

Descriptor: Attribute Format: NAMES
Template Used: N/A

Description:

LINK **Link**

Descriptor: Element Format: N/A
Template Used: N/A

Description: This element provides the capability for creating relational links with the data. It employs the HyTime "**ilink**" architectural form (template) and may contain "anchors" called (location elements) to identify two or more **linkends**. The **link** element may contain the name of the relation (e.g., **linkterm**).

LINKENDS **Link End**

Descriptor: Attribute Format: IDREFS
Template Used: N/A

Description: Used by the **link** element, this attribute contains one or more unique identifiers (IDREFs). The identifiers shall point to a CDM element or a location element which resolves at the desired data.

LINKENDTYPES **Link End Types**

Descriptor: Element Format: N/A
Template Used: N/A

Description: Specifies the type of **link** that may be used.

LOOP-NODE **Loop Node Template**

Descriptor: Entity Format: Template
Template Used: N/A.

Description: The **loop node** template allows for the creation of iterative loops within an interactive sequence (**node-seq**) of elements.

LOW-BOUND **Low Bound**

Descriptor: Element Format: N/A
Template Used: N/A.

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Description: This element is used to identify the minimum allowable entry for a numeric **fillin**.

MENU **Menu**

Descriptor: Element Format: N/A
Template Used: Node

Description: This element defines a **menu** for user interaction. It consists of a **prompt** followed by one or more **choice** elements.

MINSIZE **Minimum Size**

Descriptor: Attribute Format: NUTOKENS
Template Used: N/A

Description: Used in the **graphic** and **grphprim** elements, the **minsize** attribute specifies the minimum viewing size at which the graphic should be displayed. The minimum is expressed as the width (in inches) at which the graphic should be displayed, assuming a 36 inch viewing distance.

MODE **Mode**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Used in the **parameter** element, this attribute is composed of character data containing permissible values of either '**in**','**out**', or '**in-out**'. The default value is '**in**'. It will indicate the method of parameter passing between the technical information and the software process.

NAME **Name**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Used in all **node** elements, this attribute holds the standard nomenclature for the element expressed as character data. The permissible values of this attribute depend on the specific element type.

NAMELOC **Name Location**

Descriptor: Element Format: N/A
Template Used: N/A

Description:

NAMETYPE

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Descriptor: Attribute
Format: Character Data
Template Used: N/A

Description:

NIL **An Empty Element**

Descriptor: Element
Format: N/A
Template Used: N/A

Description: This element represents an undefined value. Any property can take on the **nil** value.

NMLIST

Descriptor: Element
Format: N/A
Template Used: N/A

Description:

NODE **Node Template**

Descriptor: Entity
Format: Template
Template Used: N/A.

Description: The **node** is a template by which technical information is defined. The **node** template contains the "content" of the technical information. The **node** template creates hierarchy within the CDM. The **node** template also contains context filtering preconditions and postconditions. The **link** element within the **node** template provides the capability to cross reference to other technical information. The use of **link**, from the Hytime model, provides additional functionality by allowing a link to be made to a document outside the CDM specification boundary.

The **node** template provides the capability to create composite structures within the content specific layer. Composite structures may contain subcomponents that employ the **node**, **node alts**, or **node seq** templates. The **node** subcomponents may be composite structures themselves or they may be primitive nodes (**text**, **tables**, **graphics**, **audio**, **video**, **process**). Composite structures create hierarchy within the CDM. When composite nodes contain other composite nodes there is an implied hierarchy. The composite node in the content model is at a lower level in the hierarchy (e.g. a **task node** contains **step nodes** in its content model).

NODE-ALTS **Node Alternatives Template**

Descriptor: Entity
Format: Template
Template Used: N/A.

Description: This template shows you how to create context sensitive filtering. This element contains one or many elements using the **node** template. **Node-alts** (node alternatives) will contain a list of mutually exclusive nodes. Their grouping is due to the fact that they apply in different contextual situations. In this manner, the **node-alts** element is a logical reference that

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contains a set of **nodes** which might apply to different situations. An important fact in the **node-alts** structure is that no hierarchy is implied between the generic identifier and the content model **nodes** (e.g. a **task-alts** element will contain **task** nodes in its content model).

NODE-SEQ **Node Sequence Template**

Descriptor: Entity Format: Template
Template Used: N/A.

Description: The **node seq** template provides the structure for creating interactive sequences with the user. The **node seq** template provides the capability to not only group elements together, but also to preserve any inherent order/sequence which may apply to the technical information. The **node seq** template also allows an author to define conditional branching and iteration within the technical information.

NUM-RANGE **Number Range**

Descriptor: Element Format: N/A
Template Used: N/A

Description: This element contains the maximum and minimum allowable values for a **fillin**.

OBNAMES

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description:

ORDERING

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description:

PARAMETER **Parameter**

Descriptor: Element Format: N/A
Template Used: N/A

Description: This element will be used to pass parameters to or from an external software **process**. For example, the 1553 bus on an aircraft might require parameters concerning a given channel which requires look up. The **parameter** element will contain the channel required by the process.

PENPATT **Pen Pattern**

Descriptor: Attribute Format: Character Date

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Template Used: N/A

Description: Used in **graphic** and **grphprim** elements, this attribute represents the bit map pattern to be used as the pen for drawing lines, points, etc. for a particular **graphic**.

PENSHAPE **Pen Shape**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Used in **graphic** and **grphprim** elements, this attribute indicates the boundary shape for the pen for drawing lines, points, etc. for a particular **graphic**.

PICID **Picture Identification**

Descriptor: Attribute Format: Number Token
Template Used: N/A

Description: Identifies which individual picture within a composite graphic the **grphprim** is referring to.

POSTCOND **Post Condition**

Descriptor: Element Format: N/A
Template Used: N/A

Description: The **postcond** element asserts the value of an **expression** to a **property** when the display system software presents a **dialog** node to the user, or when a user completes some action which needs to be recorded for later context filtering.

PRECOND **Precondition**

Descriptor: Element Format: N/A
Template Used: N/A

Description: A **precond** element must contain an expression which identifies the conditions which must be present to display the technical information.

PROCESS **External Software Process**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: This element will be used to represent an external software process.

PROCESS-ALTS **External Software Process Alternatives**

Descriptor: Element Format: N/A

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Template Used: Node, Node alts

Description: Allows for context-sensitive filtering of "PROCESS".

PROMPT **Prompt**

Descriptor: Element Format: N/A
Template Used: N/A

Description: This element defines the prompt to be displayed to the user for the presentation of a **fillin** or a **menu**. It allows the prompt to be either a text string (in the form of a question) or a graphic (a picture which requires an answer).

PROPERTY **Property**

Descriptor: Element Format: Parsable Character Data
Template Used: N/A

Description: This element contains parsable character data which represents the property (variable) name. The value of a property may be obtained by finding the current value associated with the property name in the state table.

REF **Reference**

Descriptor: Attribute Format: IDREF
Template Used: N/A

Description: Used in many elements, this attribute contains the ID of a specific element. The **ref** attribute utilizes the SGML #CONREF capability. A #CONREF attribute is only filled in when the element's content model is empty. In this case, the #CONREF attribute contains a reference which is a unique identifier to either an element employing the appropriate template or a location element that resolves to an element employing the appropriate template (see ISO/IEC IS10744).

REFTYPE **Reference Type**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description:

REMOVE **Remove**

Descriptor: Element Format: N/A
Template Used: N/A

Description: For a **set** element the **remove** element returns a **set** with **value** removed. For a **sequence** using the binary operand form it returns a **sequence** which has the first instance of **value** removed. For a **sequence** or **string** as a unary operator **remove** must contain an **index**

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value which refers to the position from which the character in the **string** is to be removed or the value in the **sequence** is to be removed. The new string or sequence will be the old one up to but not including the index position concatenating with the old one after the index position.

ROW **Row Number**

Descriptor: Attribute Format: NUTOKEN
Template Used: N/A

Description: Used in the **entry** element, the value of this attribute consists of the row number for that entry's tabular information.

ROWHDDEF **Row Header Definition**

Descriptor: Element Format: N/A
Template Used: N/A

Description: This element defines a row header for a specific row of tabular information.

SELECT **Select**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Used in the **menu** element, this attribute allows the author to designate the number of choices that may be selected by the user. The choices are either '**single**' or '**multiple**', with the default selection choice being '**single**'.

SELECTION **Selection**

Descriptor: Element Format: N/A
Template Used: N/A

Description: This element provides the capability of creating a special menu that allows selection within a given picture, text string or table.

SEQUENCE **Sequence**

Descriptor: Element Format: N/A
Template Used: N/A

Description: This element is defined as being an ordered sequence of data.

SET **Set**

Descriptor: Element Format: N/A
Template Used: N/A

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Description: This element is defined as being an unordered sequence of data.

TABLE **Table**

Descriptor: Element Format: N/A
Template Used: Node, Node-alts

Description: This element defines how a table is constructed. A **table** will contain a column header followed by one or more entries. The combination of column header and entries may be repeated for as many columns as the **table** requires.

TABLE-ALTS **Table Alternatives**

Descriptor: Element Format: N/A

Template Used: Node, Node-alts
Description: Allows for context-sensitive filtering of "TABLE".

TEXT **Text**

Descriptor: Element Format: Parsable Character Data
Template Used: Node, Node alts

Description: This element defines how text is constructed. Within a text string, there may be embedded **text** elements which allow the referencing of other elements or parts of elements through the link/location mechanism of HyTime.

TEXT-ALTS **Text Alternatives**

Descriptor: Element Format: Parsable Character Data
Template Used: Node, Node alts

Description: Allows for context-sensitive filtering of "TEXT".

TEXTCONT **Text Content**

Descriptor: Element Format: Parsable Character Data
Template Used: Node, Node alts

Description: Identifies the possible content of the **text** element.

TRANSFRM **Transformation Matrix**

Descriptor: Attribute Format: NUTOKEN
Template Used: N/A

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Description: Used in the **graphic** and **grphprim** elements, this attribute signifies a transformation matrix which specifies coordinate translations, scaling, or reflection, and rotations in terms of homogenous coordinates.

TYPE **Type**

Descriptor: Attribute
Format: Character Data
Template Used: N/A

Description: Used in all **node** elements, the information type attribute provides a more precise mechanism for classifying an element. The permissible values of this attribute are dependant upon the content specific application using this primitive.

UNOP **Unary Operator**

Descriptor: Entity
Format: N/A
Template Used: N/A

Description: This entity enumerates all of the possible unary operators which may be used within an expression. This entity could contain the following: **not**, **empty**, **size**, **head**, **tail**, **neg**, **remove**, **trunc**, **float**, **index, undef**, **max**, and **min**.

VALUE **Value**

Descriptor: Entity
Format: N/A
Template Used: N/A

Description: This entity defines an expression value. A **value** may be a **boolean**, **string**, **sequence**, **set**, **real**, **integer**, or **nil**.

VALUE-TYPE **Value Type**

Descriptor: Attribute
Format: Character Data
Template Used: N/A

Description: Used in the **property** element, this attribute is used to denote the allowable data types which may be assigned to the property. The current legal values are any combination of the following: '**boolean**', '**integer**', '**real**', '**set**', '**sequence**', '**string**', and '**general**'. The default value is '**general**'.

VIDEO **Video Sequence**

Descriptor: Element
Format: N/A
Template Used: Node, Node alts

Description: This element will be used to include a video sequence into technical information.

VIDEO-ALTS **Video Sequence Alternatives**

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Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allows for context-sensitive filtering of "VIDEO".

WINDOW **Window**

Descriptor: Attribute Format: NUTOKENS
Template Used: N/A

Description: Used in the **graphic** and **grphprim** elements, this attribute indicates the subrectangle within a graphic which should be displayed in those cases where the author wishes to display only a portion of a large graphic to the user.

X-LOCATION **X Axis Location**

Descriptor: Attribute Format: NUTOKENS
Template Used: N/A

Description: Provides the x-axis offset information.

Y-LOCATION **Y Axis Location**

Descriptor: Attribute Format: NUTOKENS
Template Used: N/A

Description: Provides the y-axis offset information.

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APPENDIX D

A CONTENT SPECIFIC LAYER TAG SET DESCRIPTIONS

D.1 SCOPE.

D.1.1 Scope. This appendix provides the detailed description of the IETM content specific elements and attributes to be included in an IETMDB. It is formulated as a description of possible tags or names for components in an IETMDB whose structure is defined by a DTD specified or developed in accordance with this specification. Unless otherwise specified by the procuring activity, this Appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

D.2 APPLICABLE DOCUMENTS.

D.2.1 Government documents.

D.2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

ISO 8879	Information Processing - Text and Office Systems - Standard Generalized Markup Language (SGML)
ISO 10744	Information Technology Hypermedia/Timebased Document Structuring Language (HyTime)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

D.3 CONTENT SPECIFIC LAYER TAG SET DESCRIPTIONS.

D.3.1 Use of SGML. The markup tags described herein conform to rules defined in ISO 8879.

D.3.2 Tag set descriptions. Data elements shall be defined in accordance with the tag set descriptions included below (see 3.1.2).

ACTION **Action**

Descriptor: Attribute
Template Used: N/A

Format: Character Data

Description: Used in the **rect** element, this attribute contains character data describing the type of maintenance action required to rectify, or fix, a fault. The action can be a '**swap**', which means it is a removal/replacement action, or it can be a '**maint**' action, which means it is an adjustment, alignment, or similar action. The default value is '**swap**'.

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AGENT **Agent**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Used in the **rect** and **test** element, this attribute contains character data describing who performs a maintenance action. It can be either a '**human**' agent, or some valid computer system (e.g., 1553 bus) called '**machine**'. The default value is '**human**'.

ALERT **Alert**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: This element identifies an alert that may accompany a task or a step. The **type** attribute may designate an alert to be a warning, caution or note which may be displayed to the technician. A warning notifies the technician that a task or step may be harmful to himself or another human if not properly performed. A caution is used in technical information to emphasize a procedure that, if not strictly followed, or a condition that, if not strictly maintained, may result in damage to the equipment. A note signifies additional information which aids the technician in completing the step or task. A note is used in technical information to emphasize an especially important procedure or condition.

ALERT-ALTS **Alert Alternatives**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allows for context-sensitive filtering of alerts.

ATTACH-PART **Attaching Part**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: This element identifies all the attaching parts required for a given part information element.

ATTACH-PART-ALTS **Attaching Part Alternatives**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allows for context-sensitive filtering of attaching parts information.

CAC **Critical Alloy Code**

Descriptor: Attribute Format: NUTOKENS

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Template Used: N/A

Description: Used in the **partbase** element, this attribute identifies the critical alloy code of an item.

CAGE **Commercial and Government Entity**

Descriptor: Attribute Format: NUTOKENS
Template Used: N/A

Description: Used in the **consum**, **equip**, and **partbase** elements, this attribute is a five character code assigned by the Defense Logistics Services Center (DLSC) to the design control activity or actual manufacturer of an item contained in the Cataloguing Handbook H4/H8 series.

CHANGENO **Change Number**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Identifies the latest change number to the revision.

CHGDATE **Change Date**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Identifies the applicable date of the latest change.

CONNECTION **Connecting Part**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: This element is used to identify a connection between two part information elements (e.g., a connection between pin 123 and wire ABC).

CONNECTION-ALTS **Connecting Part Alternatives**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allows for context-sensitive filtering of connections.

CONSUM **Consumable**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

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Description: This element identifies all the consumable required for the completion of the task.

CONSUM-ALTS **Consumable Alternatives**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allows for context-sensitive filtering of supplies/consumable.

DELETED **Deleted Data**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Indicates data deleted in this revision. There is one entry for every deleted ID.

DESIG **Designation**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Identifies the designation of the item being referenced.

DESCINFO **Descriptive Information**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: The element **descinfo** is used to define general purpose, non-procedural, narrative information such as theory of operation, schematics, etc., which are associated with a system component. The **descinfo** element is very flexible. It can be used to describe any arbitrary, hierarchical hypertext like node.

DESCINFO-ALTS **Descriptive Information Alternatives**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allows for context-sensitive filtering of descriptive information (**descinfo**).

EQUIP **Equipment**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: An **equip** element identifies the equipment needed to perform a particular **task**. **Equip** usually refers to a piece of test equipment, support equipment, or a tool.

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EQUIP-ALTS **Equipment Alternatives**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allows for context-sensitive filtering of required equipment.

ESDS **Electro-Static Discharge Sensitive Indicator**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Used in the **partbase** element, this attribute contains the electro-static discharge sensitive indicator. If an **ESDS** symbol is associated with the **partbase**, the value of this attribute should be set to '**Y3**'. If no **ESDS** symbol is associated with the **partbase**, the value of this attribute should be set to '**N3**'.

ESTTIME **Estimated Time**

Descriptor: Attribute Format: NUTOKEN
Template Used: N/A

Description: Used in the **task** and **step** elements, the value of this attribute indicates the amount of time, in minutes, required for the corresponding task/step to be completed.

EXPEND **Expendables**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Lists expendable materials used during a task.

EXPEND-ALTS **Expendables Alternatives**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allows for context-sensitive filtering of expendable materials.

FAULT **Fault**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: The element **fault** is used to identify a potential failure which may occur on a weapon system.

FAULT-ALTS **Fault Alternatives**

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Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allow for context-sensitive filtering of "FAULT".

FAULTINF **Fault Information**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: The **faultinf** element is used to define all the tests and faults associated with the system that references it.

FAULTINF-ALTS **Fault Information Alternatives**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allows context-sensitive filtering of fault information.

FAULT-TREE **Fault Tree**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: The first element in a diagnostic tree.

FAULT-TREE-ALTS **Fault Tree Alternatives**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allows for context-sensitive filtering of fault tree information.

FLTSTATE **Fault State**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: The **fltstate** element identifies a set of implicated or exculpated faults. Implicated faults are faults suspected of being bad; exculpated faults are faults known to be good. Each implicated fault will have a weight associated base on its likelihood of causing the discrepancy. The '**type**' attribute will designate whether the list of faults are '**implicated**' or '**exculpated**'.

FLTSTATE-ALTS **Fault State Alternatives**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T12:23Z
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Description: Allows for context-sensitive filtering of fault state information.

FOLLOW-ON **Follow-on Conditions**

Descriptor: Element Format: N/A
Template Used: Node Node-alts

Description: A follow-on condition is a maintenance condition which must be accomplished sometime following the completion of a task to clean up or undo actions performed during the task. For example, in order to fix a component a task might require that an access panel be removed. The panel would then need to be replaced as a follow-on action. This task might be performed sometime after the repair task is completed, but not immediately after the repair task. Other maintenance tasks might be performed in the same area before the follow-on task is accomplished. A **follow-on** element contains a set of preconditions which define the follow-on maintenance condition which must be satisfied, relational links, a text element which verbally describes the follow-on condition, a list of task(s)/step(s) which provide instructions for accomplishing the follow-on condition, and a set of post conditions which define the state changes to be made once the follow-on condition is accomplished.

FOLLOW-ON-ALTS **Follow-on Maintenance Alternatives**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allows for context-sensitive filtering of follow-on maintenance.

FSC **Federal Stock Classification**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Used in the **partbase** element, the value of this attribute contains applicable Federal Stock Classification (FSC) codes.

GOVSTD **Government Standard**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Used in the **consum** element, the value of this attribute signifies a document that establishes engineering and technical requirements for processes, procedures, practices, and methods that have been adopted as standards. It also establishes requirements for selection, application, and design criteria for materials.

HCI **Hardness Critical Item**

Descriptor: Attribute Format: Character Data
Template Used: N/A

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Description: Used in the **partbase** element, the value of this attribute represents a code which indicates that an item could degrade system survivability in a nuclear, biological, or chemically hostile environment if hardness were not considered. If an **HCI** symbol is associated with the **partbase**, the value of this attribute should be set to '**Y1**'. If no **HCI** symbol is associated with the **partbase**, the value of this attribute should be set to '**N1**'.

ICC **Item Category Code**

Descriptor: Attribute Format: Character Data
Template Used: N/A
Description: Used in the **equip** and **consum** elements, the value of this attribute signifies a code which identifies a type of item, and indicates categories into which support and test equipment, spares, repair parts, etc. may be divided.

Note: ICCs of "A," "B," and "C" should not be assigned to hardware items: these codes are reserved for grouping and selecting similar ICCs during automated data processing.

Peculiar Support Equipment and Tools not Currently in the DOD Inventory (ICC Group A):

Peculiar Support Equipment (Other)	7
Peculiar Tools	8
Peculiar Test Equipment	M
Peculiar Handling Equipment	D
Peculiar Automatic Test Equipment (ATE)	1

Common Support Equipment and Tools Currently in the DOD Inventory (ICC Group B):

Common Support Equipment (Other)	H
Common Tools	4
Common Test Equipment	5
Common Handling Equipment	6
Common Automatic Test Equipment (ATE)	2

Common Support Equipment and Tools Currently in the DOD Inventory but not Assigned to a Unit/Ship (ICC Group C):

Common Support Equipment (Other)	G
Common Tools	N
Common Test Equipment	P

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Common Handling Equipment	R
Common Automatic Test Equipment (ATE)	3
Bulk Items	Q
Training material not currently in the DOD inventory	S
Training material currently in the DOD inventory	T
End Item	W
Spare (repairable support item)	X
Repair part (a nonrepairable consumable support item, component, assembly)	Y
Repair Parts Kit	Z
A repair part, component, or assembly that is contained in a kit/set	9
Tool Kit/Set	V
Program (Embedded software)	E
Tech Manuals	F
Forms or records	J
Electrostatic Discharge-Sensitive Item	K
Electromagnetic-Sensitive Item	L
Facilities	U
System-Peculiar Spare Part	AA
Maintenance Significant Consumable	AB
Modified Hand Tool	AC
Maintenance Assist Module	AD

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IF-PARA **If Paragraphs**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allows context sensitive filtering of paragraphs based upon an "EXPRESSION".

IF-STEP **If Steps**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allows context sensitive filtering of steps based upon an "EXPRESSION".

INDEXNUM **Index Number**

Descriptor: Attribute Format: NUTOKENS
Template Used: N/A

Description: Used in the **partinfo** element, the value of this attribute contains the index number for the part which represents a call out in a graphic.

INPUT **Input Conditions**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: The **input** element contains the personnel required, the consumable used, the equipment, used and the required conditions for accomplishing a given task.

INPUT-ALTS **Input Conditions Alternatives**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allows for context-sensitive filtering of input conditions.

LOCATION **Part Location**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: The **location** element provides information for physical assessment. It will contain x, y, z location(s) for a system with respect to the x, Fuselage Station (FS), y, Buttock Line (BL), and z, Water Line (WL) reference system. Where appropriate BL may be replaced by Wing Station (WS).

LOCATION-ALTS **Part Location Alternatives**

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Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allows for context-sensitive filtering of location information.

LOCATION-X **Location X**

Descriptor: Attribute Format: NUTOKENS
Template Used: N/A

Description: Used in the **location** element, the value of this attribute contains a number representing a position on the Fuselage Station (FS), which is used as the x-axis of the weapon system.

LOCATION-Y **Location Y**

Descriptor: Attribute Format: NUTOKENS
Template Used: N/A

Description: Used in the **location** element, the value of this attribute represents a position on the Buttock Line (BL), which is used as the y-axis of the weapon system.

LOCATION-Z **Location Z**

Descriptor: Attribute Format: NUTOKENS
Template Used: N/A

Description: Used in the **location** element, the value of this attribute contains a number representing a position on the Water Line (WL), which is used as the z-axis of the weapon system.

LOOP-PARA **Loop Paragraphs**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allows for conditional repeating within paragraphs based upon an **expression**.

LOOP-STEP **Loop Step**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allows for conditional repeating within steps based upon an **expression**.

LOX **Liquid Oxygen**

Descriptor: Attribute Format: Character Data
Template Used: Node, Node alts

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Description: Used in the **partbase** element, this attribute identifies the liquid oxygen indicator. If a **lox** symbol is associated with the **partbase**, the value of this attribute should be set to '**Y2**'. If no **lox** symbol is associated with the **partbase**, the value of this attribute should be set to '**N2**'.

LRU **Line Replaceable Units**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Used in the **partinfo** element, this attribute signifies an essential support item that is removed and replaced at field level to restore the end item to its operationally ready condition. Allowable values are:

Item is a LRU	Y
Item is not a LRU	N

MAGNETIC **Magnetic Item Indicator**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Used in the **partbase** element, this attribute identifies the magnetic item indicator.

MFGCODE **Manufacturers Codes**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Used in the **consum** element, the value of this attribute indicates the in house code a manufacturer uses to represent parts.

MILSPEC **Military Specification**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Used in the **consum** element, the value of this attribute represents the exact specification for each item bought by the government.

MTBF **Mean Time Between Failure**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Used in the **fault** and **partinfo** elements, the value of this attribute signifies, for a particular interval, the total functional life of a population of an item divided by the total number of failures within the population during the measurement interval. The definition holds for time, rounds, miles, events, or other measure-of-life units.

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NSN **National Stock Number**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Used in the **equip**, **consum**, and **partbase** elements, the value of this attribute is a number, assigned under the Federal Cataloguing Program and/or North Atlantic Treaty Organization (NATO) codification of equipment system to each approved item, which provides a unique identification of an item of supply within a specified FSC. The field consists of a three character prefix, a thirteen character NSN, and a four character suffix code. For applicable codes, see DOD 4100.38-M.

OPERABILITY **Operability**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Used in the **task** element, the value of this attribute is a code used to indicate the operational status and mission readiness of the system during the maintenance task. Allowable values are:

Full Mission-Capable: performance of the maintenance task does not degrade any mission capability.	C
Partial Mission-Capable: performance of the maintenance task degrades the mission capability of the system, but can perform at least one mission.	D
System Inoperable During Equipment Maintenance: system is not available to perform all normal operations.	A
System Operable During Equipment Maintenance: system is available to perform normal operations.	B
Not Mission-Capable: system cannot perform any missions.	E
Off-Equipment Maintenance: task is performed after the item under analysis has been removed from the system.	G
Turnaround: task occurs during normal turnaround operations, and does not affect the operability of the system.	F

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OUTCOME **Outcome**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: This element represents a possible outcome from a test. It identifies a fault state (**fltstate**) for use in a dynamic fault model, and a **test** or **fault** for the static tree model.

OUTCOME-ALTS **Outcome Alternatives**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allows for context-sensitive filtering of test outcomes.

PARA **Para**

Descriptor: Element Format: N/A
Template Used: Node, Node alts, Node seq, If node, Loop node

Description: Identifies a piece of text to be displayed under the **descinfo** element. The text may be "Theory of Operation", "General Information", etc.

PARA-ALTS **Para Alternatives**

Descriptor: Element Format: N/A
Template Used: Node, Node alts, Node seq, If node, Loop node

Description: Allows context-sensitive filtering of paragraphs.

PARA-SEQ **Paragraph Sequence**

Descriptor: Element Format: N/A
Template Used: Node, Node alts, Node seq, If node, Loop node

Description: Allows for paragraphs to be arranged in sequences.

PARTBASE **Part Base**

Descriptor: Element Format: N/A
Template Used: Node Node-alts

Description: This element describes the supply system's view of the part information. It describes the item in terms of its part number.

PARTBASE-ALTS **Base Supply Parts Information Alternatives**

Descriptor: Element Format: N/A

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Template Used: Node Node-alts

Description: Allows for context-sensitive filtering of Base Supply parts information.

PARTINFO **Part Information**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: This element describes the maintainer's view of the part information. It identifies parts information within its relative position in the weapon system.

PARTINFO-ALTS **Part Information Alternatives**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allows for context-sensitive filtering of parts information.

PARTNUM **Part Number**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Used in the **partbase** element, this attribute signifies any number, other than a government activity stock number, used to identify an item of production or supply.

PERSON **Person**

Descriptor: Element Format: N/A
Template Used: Node Node-alts

Description: This element is used to identify the personnel requirements for a given task.

PERSON-ALTS **Person Alternatives**

Descriptor: Element Format: N/A
Template Used: Node Node-alts

Description: Allows for context-sensitive filtering of personnel information.

PMIC **Precious Metal Indicator Code**

Descriptor: Attribute Format: NUTOKEN
Template Used: N/A

Description: Used in the **partbase** element, this attribute identifies the precious metal indicator code.

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QEC **Quick Engine Change**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Used in the **partbase** element, this attribute identifies the quick engine change indicator. If a **qec** symbol is associated with the **partbase**, the value of this attribute should be set to '**Y4**'. If no **qec** symbol is associated with the **partbase**, the value of this attribute should be set to '**N4**'.

QUANTITY **Quantity**

Descriptor: Attribute Format: CDATA
Template Used: N/A

Description: Used in the **person**, **equip**, **expend**, and **consum** elements, the value of this attribute signifies the amount of the appropriate consumable, equipment, or people required for the associated **task**/**step**.

QPEI **Quantity Per End Item**

Descriptor: Attribute Format: NUTOKEN
Template Used: N/A

Description: Used in the **partbase** element, this attribute identifies the quantity per end item used.

RANGE **Range**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Used in the **test** element, this attribute represents the boundaries for valid choices or outcomes, according to the element containing the range.

RECT **Rectification**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: The **rectification** element identifies the prescribed task that will repair the fault causing the discrepancy and all other faults that could be fixed by the rectification. Upon completion of the task, a test is performed to verify the effect of the rectification.

RECT-ALTS **Rectification Alternatives**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

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Description: Allows context-sensitive filtering of rectification (**rect**).

REFDES **Reference Designation**

Descriptor: Attribute Format: NUTOKEN
Template Used: N/A

Description: Used in the **partinfo** element, this attribute is an identifier assigned according to a numbering scheme for parts of a system which reflects the hierarchical assembly of the system.

REFMAT **Reference Material**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: This element is used to support inclusions of reference material.

REFMAT-ALTS **Reference Material Alternatives**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allows for context-sensitive filtering of reference material (**refmat**).

REPLVL **Replenishment Level**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Used in the **partinfo** element, this attribute represents the minimum quantity of a part in stock that will trigger a reorder or stock action.

REVISION **Revision**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Identifies a unique alpha identifier for each **revision**.

REVDATE **Revision Date**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Identifies the applicable date of the revision.

REQCOND **Required Conditions**

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Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: A **reqcond** element contains a list of preliminary conditions which must be met prior to beginning a **task**. If any condition is not met, it contains the **task** or **step** which will satisfy the condition. It also contains **post condition**s which will record the state changes made in satisfying the conditions.

REQCOND-ALTS **Required Condition Alternatives**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allows for context-sensitive filtering of required conditions.

SERVICEDES **Service Designator**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Used in the **task** element, this attribute is a single position code identifying the military service or nonmilitary major governmental agency having jurisdiction over, or executive management responsibility for, the acquisition. Allowable values are:

Army	A
Air Force	F
Marine Corps	M
Navy	N
Coast Guard	Y
All Military	X
Federal Aviation Administration	T
FAA/All Military	J
National Security Agency	S
Other	O

SMR **Source, Maintenance, Recoverability Codes**

Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: Used in the **partbase** element, SMR codes are alphabetic or alphanumeric symbols used at the time of provisioning to indicate the source of supply of an item, its maintenance implications, and its recoverability characteristics. The provisioning activity may require the contractor to recommend these codes. Approved codes are defined in: AR 700-82, OPNAVINST 4410.2, AFR 66-45, MCO 4400.120, and DSAR 4100.6.

STATUS **Status**

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Descriptor: Attribute Format: Character Data
Template Used: N/A

Description: In conjunction with **version,** this attribute indicates updated (changed), added (new), or deleted data.

STEP **Step**

Descriptor: Element Format: N/A
Template Used: Node, Node alts, Node seq, If node, Loop node

Description: The **step** element is the primary component of a maintenance procedure. It describes the actions to be performed in order to successfully complete a task.

STEP-ALTS **Step Alternatives**

Descriptor: Element Format: N/A
Template Used: Node, Node alts, Node seq, If node, Loop node

Description: Allows for context-sensitive filtering of a **step**.

STEP-SEQ **Step Sequence**

Descriptor: Element Format: N/A
Template Used: Node, Node alts, Node seq, If node, Loop node

Description: Identifies the sequence of steps.

SYSTEM **System**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: The **system** element defines the vehicle/system/subsystem/subassembly hierarchy for the weapon system. A **system** element must be created for any component (ie., vehicle, system, subsystem, subassembly) which has associated technical information (ie., descriptive, procedural, fault, or part information).

SYSTEM-ALTS **System Alternatives**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allows for context-sensitive filtering of the **system**.

TASK **Task**

Descriptor: Element Format: N/A

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Template Used: Node, Node alts

Description: The **task** element is a set of directive steps which make up a specific maintenance procedure. A maintenance procedure could be a preventive or corrective maintenance task. Preventive tasks are performed at regular intervals to ensure that the item or system will continue to operate correctly and safely (such as inspect, clean, lubricate, etc). Corrective (or unscheduled) maintenance procedures are performed when required to repair faulty items or systems that have been identified by troubleshooting procedures. A procedural task is made up of steps, and ties all text, graphics, messages, prompts, and references required to convey the step together. A **task** element contains linking information necessary to link one **task** to other **task**s.

TASK-ALTS **Task Alternatives**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allows for context-filtering of a **task**.

TECHINFO **Technical Information**

Descriptor: Element Format: N/A
Template Used: N/A

Description: This element represents the top layer of the information contained in this content specific DTD. The content model contains the top level system such as "F-15", "M-1", or "F/A-18".

TEST **Test**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: This element indicates a diagnostic test that will lead to outcomes and guide the technician toward a rectification during troubleshooting.

TEST-ALTS **Test Alternative**

Descriptor: Element Format: N/A
Template Used: Node, Node alts

Description: Allows for the context-sensitive filtering of a diagnostic **test**.

UNIT-OF-MEASURE **Unit of Measure**

Descriptor: Attribute Format: Character Data
Template Used: N/A

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Description: Used in the **consum** element, this attribute identifies the type of unit measurement used to quantify the number of consumables needed for the current application. (e.g., "inches", "meters", "pounds", etc.).

UNITSPER **Units per Assembly, System, etc.**

Descriptor: Attribute Format: NUTOKEN
Template Used: N/A

Description: Used in the **partinfo** element, this attribute represents the number of units required per assembly of a system or component.

USABLEON **Usable On Code**

Descriptor: Attribute Format: NUTOKEN
Template Used: N/A

Description: Used in the **partinfo** element, this attribute identifies the different configurations in which a part or assembly may appear within a system or vehicle.

VERSION **Version**

Descriptor: Element Format: N/A
Template Used: Node

Description: This element identifies the current version of the data by providing the last revision information and change information necessary for taking delivery of partial databases.

WEIGHT **Fault Probability**

Descriptor: Attribute Format: NUTOKENS
Template Used: N/A

Description: Used in the **fltstate** element, this attribute represents a probability associated with a given **fault** within a list of **fault**s in a fault state (**fltstate**).

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-87269A	2. DOCUMENT DATE (YYMMDD) 95/10/01

3. DOCUMENT TITLE
Data Base, Revisable - Interactive Electronic Technical Manuals, For the Support of

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	e. DATE SUBMITTED *(YYMMDD)*

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a. NAME Steven C. Holloway	b. TELEPHONE *(Include Area Code)* (1) Commercial: (513) 427-5869 (2) AUTOVON: 787-5869
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